

Project: Identifying Risk Factors of Diabetes Using BRFSS 2015 Data

**Leona Lu (278049476)
Faculty of Science, University of Auckland
August 2025**

Table of Contents

Project: Identifying Risk Factors of Diabetes Using BRFSS 2015 Data	1
Step 1: Business and Situation Understanding	3
1.1 Identify the Objectives of the Business/Situation	3
1.2 Assess the Situation	4
1.3 Determine Data Mining Objectives	4
1.4 Produce a Project Plan	5
Step 2: Data Understanding	5
2.1 Collect Initial Data	5
2.2 Describe the Data	6
2.3 Explore the Data	7
2.4 Verify the Data Quality	10
Step 3: Data Preparation	11
3.1 Data Selection	11
3.2 Data Cleaning	12
3.3 Data Construction	13
3.4 Data Integration	15
3.5 Data Reformatting	15
Step 4: Data Transformation	16
4.1 Data Reduction	16
4.2 Data Projection	19
Step 5: Data Mining Methods Selection	20
5.1 Match and discuss DM methods within the context of the DM objectives	20
5.2 Select the Appropriate DM Method(s)	21
Step 6: Data Mining Algorithms Selection	21
6.1 Exploratory Analysis of DM Algorithms within the context of the DM objectives	21
6.2 Select the Appropriate DM Algorithms	25
6.3 Model Building and Parameter Selection	25
Step 7: Data Mining	27
7.1 Test Design	27
7.2 Data mining Execution	28
7.3 Search for patterns and document the model's output	28
Step 8: Interpretation	43
8.1 Study and Discuss the Mined Patterns	43
8.2 Interpretation of Results, Models and Patterns	44
8.3 Cluster Analysis for Health Profile Segmentation	48
8.4 Evaluation of Results, Models and Patterns	50
8.5 Multiple Iterations and Model Refinement	51
References	53
Academic Integrity Disclaimer	53

Step 1: Business and Situation Understanding

1.1 Identify the Objectives of the Business/Situation

This project aligns with Sustainable Development Goal (SDG) 3.4, which aims to reduce premature mortality from non-communicable diseases (NCDs), including diabetes, by one-third by 2030. Globally, diabetes is recognized as one of the fastest-growing health emergencies of the 21st century, with projections indicating that the number of adults with diabetes could reach to 853 million by 2050 (*International Diabetes Federation, 2025*).

In the United States, the scale of the issue is particularly acute. According to the CDC's National Diabetes Statistics Report, an estimated 38.4 million Americans (11.6% of the population) had diabetes in 2021, with a staggering 8.7 million of those cases going undiagnosed (*Centers for Disease Control and Prevention, 2024*). This epidemic carries a monumental economic burden, with the total estimated cost of diagnosed diabetes in the U.S. reaching \$412.9 billion in 2022, comprising \$306.6 billion in direct medical costs and \$106.3 billion in reduced productivity (*American Diabetes Association, 2023*).

Given that nearly a quarter of adults with diabetes are unaware of their condition (*Centers for Disease Control and Prevention, 2024*), early identification of at-risk individuals is a critical public health imperative. Research has shown that early detection and subsequent treatment can lead to significant long-term cost savings for healthcare systems (*Sortsø et al., 2021*). In fact, a major Danish population study demonstrated that for individuals found to have diabetes, a screening-based detection approach was associated with significantly lower healthcare costs compared to routine-care detection, with the cost difference being offset within two years of the screening program (*Sortsø et al., 2021*). Therefore, the development of effective predictive tools is not just a clinical goal but an economic necessity.

Objective:

To develop a predictive model that identifies adults at high risk of diabetes using behavioral and physiological health indicators from the BRFSS 2015 dataset. This model can support public health agencies in designing targeted screening programs and preventive campaigns, and at the same time provide actionable recommendations for individuals to reduce the risk of diabetes.

Business/Situation Success Criteria:

- Enable early detection to reduce complications (e.g., kidney failure, blindness).
- Support resource allocation by identifying top modifiable risk factors.
- Achieve model performance that is both accurate and interpretable for healthcare professionals.

1.2 Assess the Situation

Aspect	Description
Resources	IBM SPSS Modeler v18.4 (University-licensed), diabetes_binary_health_indicators_BRFSS2015.csv" dataset (253,680 records, 22 variables), SPSS tutorials, and computing facilities.
Requirements	Binary classification model with $\geq 80\%$ accuracy and $\geq 75\%$ recall; feature importance analysis; reproducible workflow.
Assumptions	The dataset is representative of U.S. adult population; self-reported data is reasonably accurate; no temporal drift in health behaviors since 2015.
Constraints	Data is U.S.-only and self-reported; limited to cross-sectional analysis; no real-time deployment in this iteration.
Risks	Class imbalance (84.9% non-diabetic); overfitting due to high dimensionality; bias from self-reporting (e.g., underreporting of alcohol use).
Contingencies	Use SMOTE(Synthetic Minority Over-sampling Technique) or undersampling to balance classes; apply regularization in logistic regression; perform sensitivity analysis by excluding potentially biased variables.

1.3 Determine Data Mining Objectives

The data mining goals are defined using the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound):

1. To build a binary classification model using SPSS Modeler that predicts Diabetes_binary (0 = No, 1 = Yes) with:

- Accuracy $\geq 80\%$
- Recall (Sensitivity) $\geq 75\%$
- AUC ≥ 0.85

on a 30% holdout test set.

2. To identify the top 5 most influential predictors of diabetes using feature importance from Random Forest or logistic regression coefficients.

3. To evaluate the impact of data balancing techniques (e.g., SMOTE vs. undersampling) on model performance, measured by changes in recall and F1-score.
4. To translate model insights into actionable recommendations for individuals and public health campaigns. For example, quantifying the risk reduction associated with achieving a healthy weight or increasing physical activity.

These goals are directly aligned with the business objective of early detection and public health intervention.

1.4 Produce a Project Plan

Dates	Step	Task Description
Aug 1–2, 2025	BU	Finalize business objectives, success criteria, risks, and assumptions. Link to SDG 3.4.
Aug 3–4, 2025	DU	Import dataset into SPSS; generate summary statistics; create histograms, bar charts, and boxplots; check class distribution.
Aug 5–6, 2025	DP	Assign data types; handle outliers (e.g., BMI > 60); ensure no missing values; derive new features (e.g., BMI category).
7-Aug-25	DT	Normalize BMI; apply log transformation if skewed; encode categorical variables (e.g., GenHlth).
8-Aug-25	DMM	Confirm classification as the data mining task.
9-Aug-25	DMA	Select Random Forest and Logistic Regression for comparison.
Aug 10–12, 2025	DM	Train models on 70% training set; evaluate on 30% test set; apply SMOTE for class balancing.
Aug 13–14, 2025	INT	Interpret results; visualize confusion matrices, ROC curves, and feature importance; iterate model tuning.
Aug 15–16, 2025	ACT	Summarize findings; compare with medical literature; draft recommendations for public health use.
Aug 17–18, 2025	Finalization	Compile report, appendices, and SPSS flow; proofread and submit.

Step 2: Data Understanding

2.1 Collect Initial Data

- Source: The dataset diabetes_binary_health_indicators_BRFSS2015.csv was downloaded from Kaggle (<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>).
- Origin: Collected by the U.S. Centers for Disease Control and Prevention (CDC) through the Behavioral Risk Factor Surveillance System (BRFSS) 2015, a cross-sectional telephone survey of U.S. adults.
- Collection Method: Self-reported responses to standardized health questions.
- Issues Encountered: No access or ethical issues; data is publicly available and anonymized. However, self-reporting may introduce recall or social desirability bias (e.g., underreporting of alcohol use or BMI).

2.2 Describe the Data

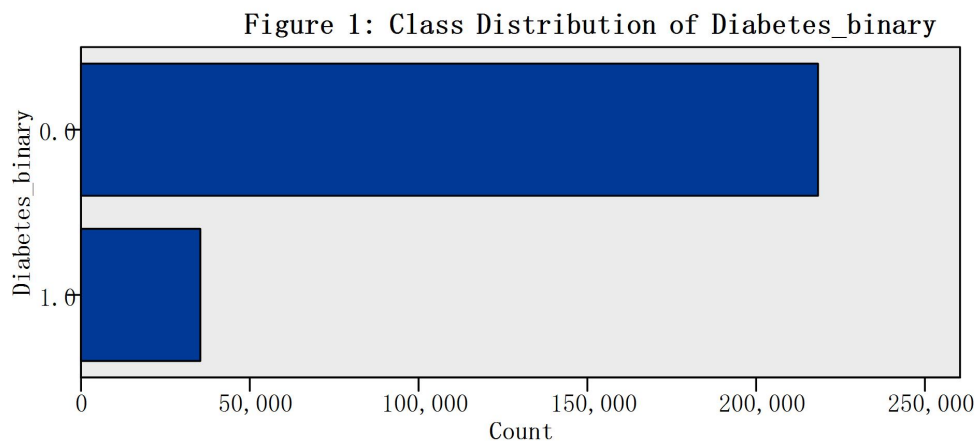
Attribute	Type	Description
Diabetes_binary	Binary	Target: 1 = Diabetic, 0 = Non-diabetic
HighBP	Binary	0 = no high Blood Pressure, 1 = high BP
HighChol	Binary	1 = High cholesterol, 0 = Normal
CholCheck	Binary	1 = Cholesterol checked in past 5 years
BMI	Continuous	Body Mass Index (kg/m ²)
Smoker	Binary	1 = Smoked ≥100 cigarettes in entire life
Stroke	Binary	1 = History of stroke
HeartDiseaseorAttack	Binary	1 = Coronary heart disease or heart attack
PhysActivity	Binary	1 = Physical activity in past 30 days
Fruits	Binary	1 = Consume fruit ≥1/day
Veggies	Binary	1 = Consume vegetables ≥1/day
HvyAlcoholConsump	Binary	1 = Heavy drinkers (adult men >14 drinks/week, women >7)
GenHlth	Ordinal (1–5)	1 = Excellent, 2 = Very good, 3 = Good, 4 = Fair, 5 = Poor
MentHlth	Integer (0–30)	Days mental health not good
PhysHlth	Integer (0–30)	physical illness or injury days in past 30 days scale 1-30
DiffWalk	Binary	having serious difficulty walking or climbing stairs
Sex	Binary	1 = Male, 0 = Female
Age	Ordinal (1–13)	13-level age category (1 = 18-24, 9 = 60-64, 13 = 80 or older)
Education	Ordinal (1–6)	1 = Never attended school or only kindergarten, 2 = Grades 1 through 8 (Elementary) 3 = Grades 9 through 11 (Some high school)

		4 = Grade 12 or GED (High school graduate)
		5 = College 1 year to 3 years (Some college or technical school)
		6 = College 4 years or more (College graduate)
Income	Ordinal (1–8)	1 = <\$10k,
		2 = [\$10k, \$15k),
		3 = [\$15k, \$20),
		4 = [\$20k, \$25k),
		5 = [\$25k, \$35k),
		6 = [\$35k, \$50k),
		7 = [\$50k, \$75k),
		8 = ≥\$75k

- Total Records: 253,680
- Target Distribution: 86.1% (218,334) non-diabetic, 13.9% (35,346) diabetic → Class imbalance present.

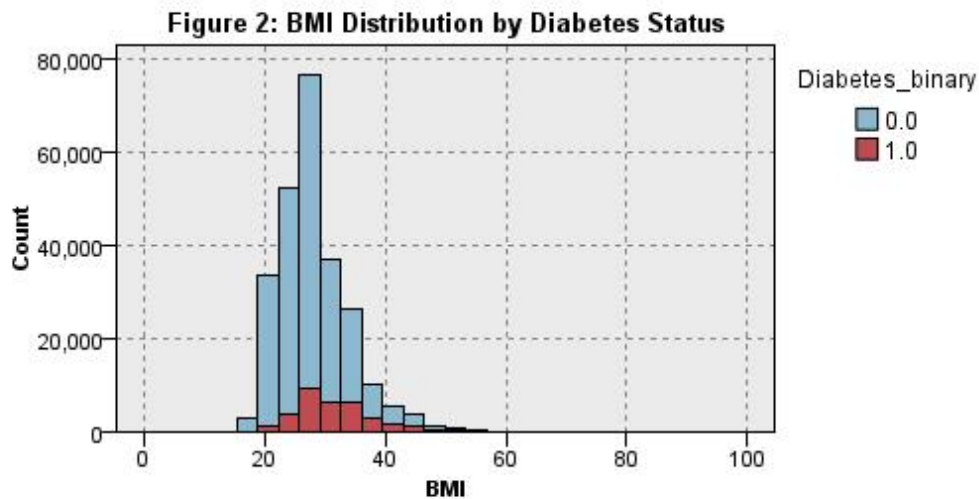
2.3 Explore the Data

Data exploration was conducted using visualizations in SPSS Modeler to understand distributions and relationships.

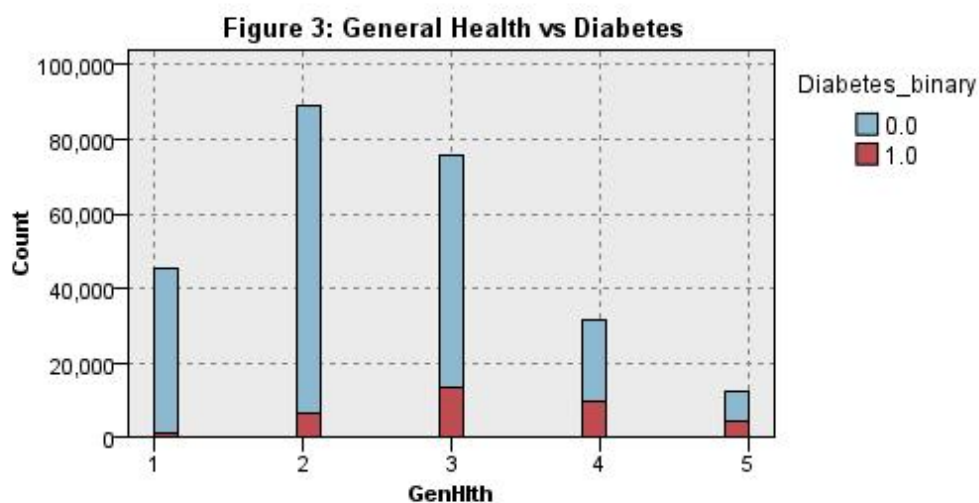


Diabetes_binary	__count__	V3
0.0	218334.0022	Diabetes_binary = 0.0 Count = 218334
1.0	35346.0006	Diabetes_binary = 1.0 Count = 35346

Description: Only 13.93% of the sample has diabetes, indicating class imbalance. This justifies the need for SMOTE or weighting in modeling.



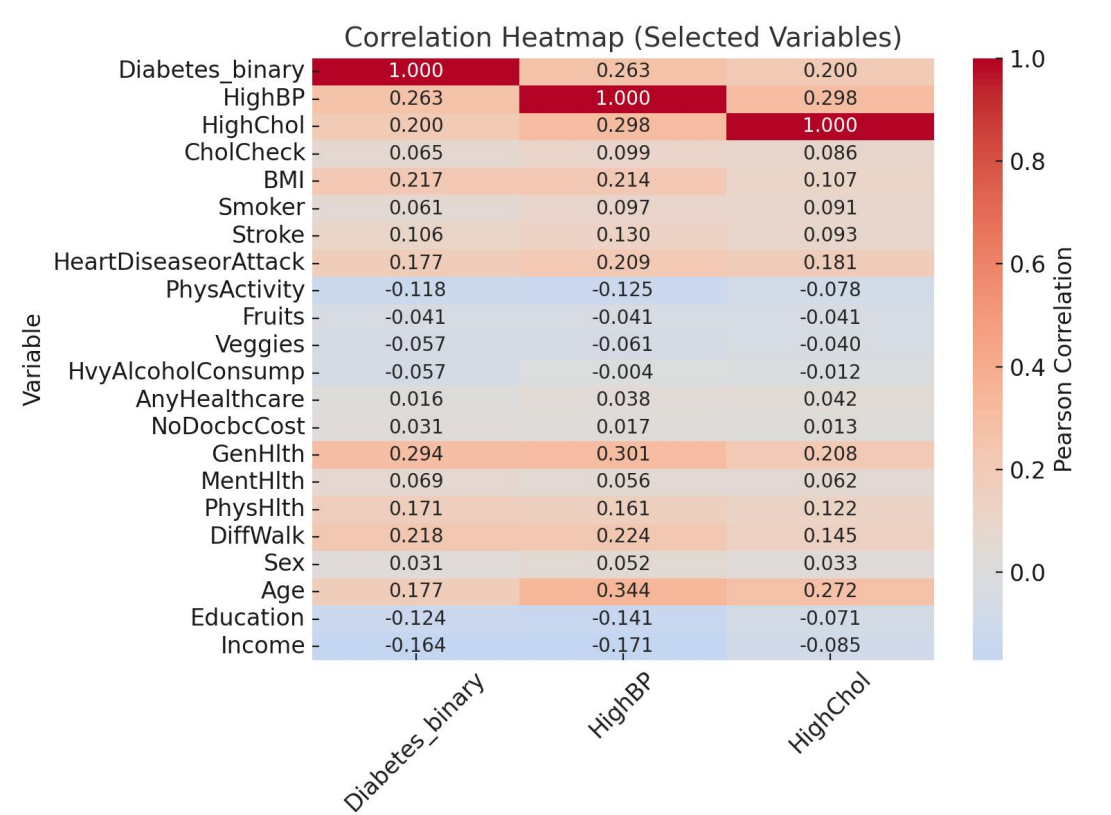
Description: This chart shows how Body Mass Index (BMI) is distributed for people with and without diabetes. The blue bars represent people without diabetes, and the red bars represent people with diabetes. The graph clearly shows that people with diabetes tend to have higher BMIs. The peak for the diabetic group (red) is around a BMI of 30, which is in the "obese" range, while the peak for the non-diabetic group (blue) is lower, around a BMI of 25, which is in the "overweight" range. This indicates a strong link between higher body weight and the risk of diabetes.



Description: This bar chart illustrates the distribution of self-reported general health status (GenHlth, ranging from 1 = Excellent to 5 = Poor) among individuals with and without diabetes in the dataset. Each stacked bar represents the count of participants for a given health rating, with blue segments (0.0) indicating non-diabetic individuals and red segments (1.0) indicating diabetic individuals.

The chart shows that poorer general health ratings (higher GenHlth values) are associated with a higher proportion of diabetic individuals. While most participants report good to fair health (ratings 2 and 3), the proportion of diabetics increases notably as general health worsens, particularly at ratings 4 and 5.

Figure 4: Correlation Heatmap (All Variables)



High positive correlations: GenHlth & HighBP

Description: This Correlation Heatmap displays the Pearson correlation coefficients between the target variable Diabetes_binary and all other variables in the dataset. The values range from -1 to +1, where positive values indicate a direct relationship (e.g., higher BMI tends to be associated with higher diabetes risk), and negative values indicate an inverse relationship. The strength of the correlation is indicated by the label "Strong" for values above $|0.2|$, which is a commonly used heuristic in social and health sciences to indicate a practically significant, albeit not powerful, relationship. As shown, several variables exhibit strong positive correlations with diabetes status, including HighBP (0.263), HighChol (0.200), BMI (0.217), GenHlth (0.294), and DiffWalk (0.218). This suggests that **high blood pressure, high**

cholesterol, obesity, poor self-rated general health, and difficulty walking are significant risk factors for diabetes. Variables like PhysActivity (-0.118) show a weak negative correlation, indicating physical activity may be protective.

2.4 Verify the Data Quality

The data quality was rigorously verified using IBM SPSS Modeler's built-in "Data Audit" node to systematically identify potential errors, missing values, and data quality patterns. The results are summarized below:

Missing Values: The audit confirmed that the dataset is complete with no missing values across all 22 variables. This indicates high data integrity in terms of completeness and eliminates the need for imputation.

Outliers and Extreme Values: While there are no invalid entries (e.g., negative BMI), several extreme values were identified in the continuous variables:

* **BMI:** The maximum value was found to be 96.28, which is physiologically implausible for a living adult and is likely a data entry error. Similarly, values above 60 are highly suspicious. A total of 385 records had a BMI greater than 60.

* **MentHlth & PhysHlth:** The maximum value for both MentHlth and PhysHlth is 30, which represents the full 30 days of the month. While this is not an error, it indicates that some individuals reported poor mental or physical health every day of the past month, which is a valid but extreme case.

Data Consistency and Types: All variables were found to be within logical ranges (e.g., Diabetes_binary only contains 0 and 1, GenHlth ranges from 1 to 5). The data types were correctly assigned as numeric, with appropriate measurement levels (nominal, ordinal, scale) for modeling.

Data Quality Pattern: The most significant data quality pattern observed is the severe class imbalance in the target variable Diabetes_binary. The audit confirmed that 86.1% of the records (218,334 individuals) are non-diabetic (0), while only 13.1% (35,346 individuals) are diabetic (1). This imbalance is a critical quality consideration for model training, as it can lead to biased models that perform poorly on the minority class.

Figure 5: Screenshot of SPSS Variable Quality

Complete fields (%): 100% Complete records (%): 100%

Field	Measurement	Outliers	Extremes	Action	Impute Missing	Method	% Com...	Valid Records	Null Value	Empty String
Diabetes_bi...	Flag	--	--	Never	Fixed		100	253680	0	0
HighBP	Flag	--	--	Never	Fixed		100	253680	0	0
HighChol	Flag	--	--	Never	Fixed		100	253680	0	0
CholCheck	Flag	--	--	Never	Fixed		100	253680	0	0
BMI	Continuous	2193	770	None	Never	Fixed	100	253680	0	0
Smoker	Flag	--	--	Never	Fixed		100	253680	0	0
Stroke	Flag	--	--	Never	Fixed		100	253680	0	0
HeartDiseas...	Flag	--	--	Never	Fixed		100	253680	0	0
PhysActivity	Flag	--	--	Never	Fixed		100	253680	0	0
Fruits	Flag	--	--	Never	Fixed		100	253680	0	0
Veggies	Flag	--	--	Never	Fixed		100	253680	0	0
HvyAlcoholC...	Flag	--	--	Never	Fixed		100	253680	0	0
AnyHealthcare	Flag	--	--	Never	Fixed		100	253680	0	0
NoDocbcCost	Flag	--	--	Never	Fixed		100	253680	0	0
GenHlth	Ordinal	--	--	Never	Fixed		100	253680	0	0
MentHlth	Continuous	12697	0	None	Never	Fixed	100	253680	0	0
PhysHlth	Continuous	0	0	None	Never	Fixed	100	253680	0	0
DiffWalk	Flag	--	--	Never	Fixed		100	253680	0	0
Sex	Flag	--	--	Never	Fixed		100	253680	0	0
Age	Ordinal	--	--	Never	Fixed		100	253680	0	0
Education	Ordinal	--	--	Never	Fixed		100	253680	0	0
Income	Ordinal	--	--	Never	Fixed		100	253680	0	0

Conclusion: The dataset is clean in terms of missing data and categorical consistency. However, the presence of extreme BMI values and the significant class imbalance are the primary data quality issues that will be addressed in the subsequent Data Preparation (Step 3) and Data Mining (Step 7) phases, respectively. The extreme BMI values will be capped, and the class imbalance will be mitigated using resampling techniques like SMOTE.

Step 3: Data Preparation

Based on the data quality assessment from Step 2, the dataset was systematically prepared for modeling by addressing identified issues and structuring the data appropriately. All operations were executed using IBM SPSS Modeler.

3.1 Data Selection

The initial dataset comprised 22 variables across 253,680 observations. Variable selection was explicitly guided by the data mining goal (predicting diabetes status), data quality patterns, and technical constraints.

Target Variable: Diabetes_binary was confirmed as the binary target for classification.

Inclusion Criteria: Variables with established physiological or behavioral relevance to diabetes risk, such as **HighBP**, **HighChol**, **BMI**, **GenHlth**, **DiffWalk**, and **Age**, were retained as core predictors.

Exclusion Criteria: Two variables, **AnyHealthcare** and **NoDocbcCost**, were excluded. This decision was not based on low correlation, but on the critical principle of

preventing data leakage. These variables represent access to healthcare, which is more likely a consequence of having diabetes (or being at high risk) rather than a cause. Including them could artificially inflate model performance and obscure the identification of genuine, modifiable risk factors, which is central to the project's public health goal

Technical Constraints: The dataset size is computationally manageable. However, the significant class imbalance (86.1% non-diabetic) remains a key constraint, to be addressed in Step 7 via algorithmic balancing.

3.2 Data Cleaning

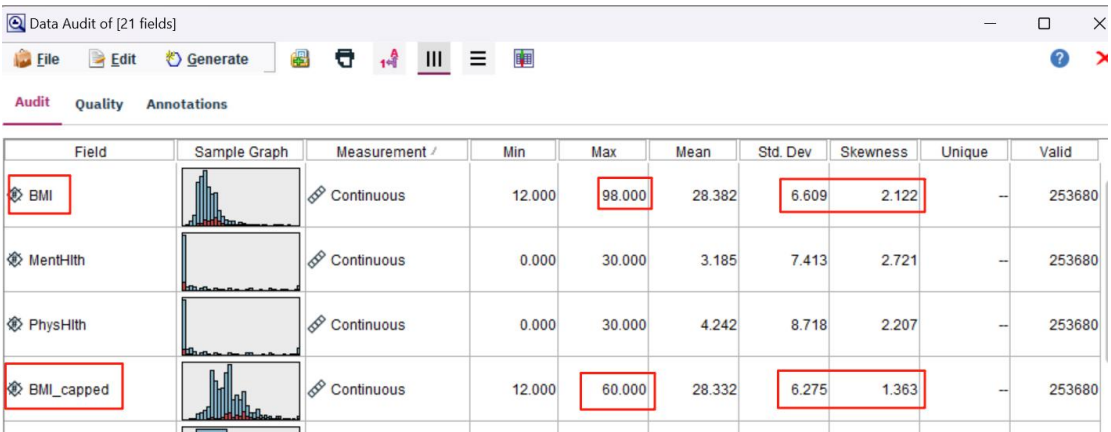
This step explicitly addressed the data quality issues identified in Step 2.4.

Missing Values: The "Data Audit" node confirmed 0% missing values across all variables. Consequently, no imputation or deletion was required, ensuring full utilization of the dataset.

Handling Outliers (Extreme Values)

- **Issue:** The BMI variable contained physiologically implausible values (e.g., 96.28), which could skew model training.
- **Resolution:** Using the **Derive** node, a capping rule was applied: if (BMI > 60) then 60 else BMI endif .
- **Impact:** This adjustment modified **805 records** (0.32% of the dataset), reducing the BMI standard deviation from **6.609** to **6.275** without significantly altering the mean, thus maintaining data integrity while reducing the influence of extreme variance.

Figure 6: Screenshot of SPSS Audit After Handling BMI Extremes



Logical Consistency: All categorical and ordinal variables (e.g., Diabetes_binary, GenHlth) were verified to be within their defined ranges. No recoding or correction was necessary for consistency.

3.3 Data Construction

New features were constructed using the "Derive" node to create more powerful, interpretable, and non-linear predictors.

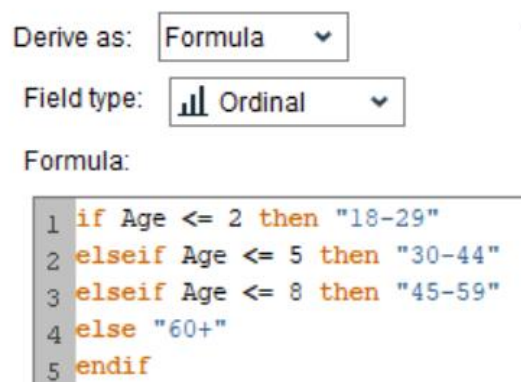
BMI_Category (Nominal):

- Rationale: To capture the non-linear relationship between BMI and diabetes risk and to improve model interpretability.
- Construction: BMI was binned into categories based on WHO standards: **Underweight** (BMI < 18.5), **Normal** (18.5 ≤ BMI < 25), **Overweight** (25 ≤ BMI < 30), and **Obese** (BMI ≥ 30).

Age_Group (Ordinal):

- Rationale: To model potential non-linear age effects and facilitate the analysis of interactions between age and other risk factors.
- Construction: The original 13-level Age variable was mapped to broader, more interpretable brackets: 18–29, 30–44, 45–59, and 60+.

Figure 7: Screenshot of SPSS Health_Index Derive Age_Group



While conceptually promising, these derived features did not yield the expected improvements in predictive performance. This suggests that automatic feature selection within tree-based models may already capture the relevant non-linearities, making such manual transformations less effective in this context

Health_Index (Continues):

- Rationale: To create a single, normalized measure of overall health burden by combining self-reported **general health** and recent **physical/mental health days**. The smaller numbers in the measure shall represent a better overall health state.
- Construction: The variables were normalized to a 0-1 scale using min-max scaling, and their average was calculated.

$$((\text{GenHlth} - 1) / 4)$$

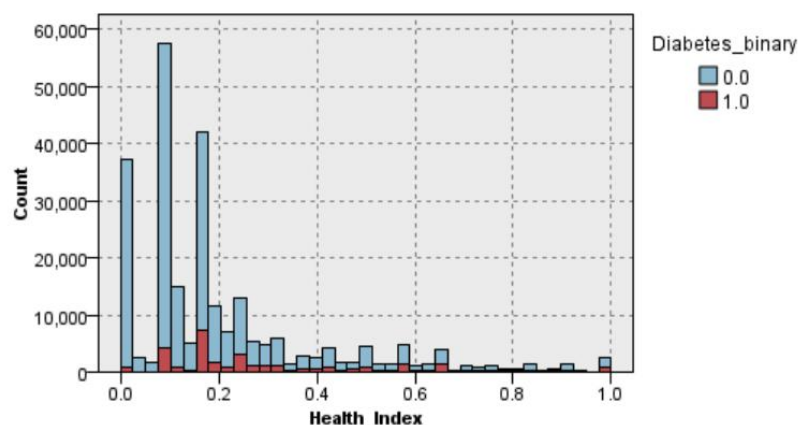
$$+ (\text{PhysHlth} / 30)$$

$$+ (\text{MentHlth} / 30)$$

$$) / 3$$

Outcome and Interpretation: The resulting Health_Index did not exhibit the expected relationship with diabetes status. As shown in Figure 7, the distribution of non-diabetic individuals is concentrated at lower Health_Index values, but the proportion of diabetic individuals does not increase monotonically with higher Health_Index. This suggests that the combination of these three variables may not be an effective measure of overall health burden for this specific task.

Figure 8: Screenshot of SPSS Health_Index Histogram



Validation of Health_Index: Correlation analysis revealed that Health_Index had only a weak-to-moderate correlation with diabetes status ($r = 0.233$), which is lower than the direct correlation of GenHlth ($r = 0.294$). Furthermore, Health_Index was highly correlated with its component variables (GenHlth, $r = 0.792$; PhysHlth, $r = 0.831$; MentHlth, $r = 0.690$), indicating strong redundancy. This suggests that aggregating these variables into a composite index did not add predictive value and may introduce multicollinearity in modeling. Based on this evidence, Health_Index

was excluded from the final model, while GenHlth and PhysHlth were retained as independent predictors.

Figure 9: Screenshot of SPSS Health_Index Pearson Correlations

Pearson Correlations

	Health_Index	Diabetes_binary	GenHlth	MentHlth	PhysHlth
Health_Index	1.000/Perfect	0.233/Strong	0.792/Strong	0.690/Strong	0.831/Strong
Diabetes_binary	0.233/Strong	1.000/Perfect	0.294/Strong	0.069/Strong	0.171/Strong
GenHlth	0.792/Strong	0.294/Strong	1.000/Perfect	0.302/Strong	0.524/Strong
MentHlth	0.690/Strong	0.069/Strong	0.302/Strong	1.000/Perfect	0.354/Strong
PhysHlth	0.831/Strong	0.171/Strong	0.524/Strong	0.354/Strong	1.000/Perfect

3.4 Data Integration

Assessment: The project is based on a single, unified source: the diabetes_binary_health_indicators_BRFSS2015.csv dataset.

Action: No external data integration or merging was required for this iteration. Metadata from the CDC documentation was used solely for variable interpretation and was not merged into the analytical dataset.

3.5 Data Reformatting

Measurement Level Assignment: Using the *Type* node, each variable was explicitly assigned to its correct measurement level:

Nominal (Categorical): Diabetes_binary, HighBP, HighChol, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, DiffWalk, Sex, Obese, BMI_Category.

Ordinal: Age_Group, GenHlth, Education, Income.

Scale (Continuous): BMI_capped, MentHlth, PhysHlth, RiskScore.

Target Variable Confirmation: *Diabetes_binary* was formally designated as the target field for supervised classification tasks.

Format Harmonization: Categorical variables were checked to ensure binary coding (0 = No, 1 = Yes) was consistent across the dataset. Ordinal variables (e.g., Age_Group, GenHlth) were confirmed to have values in ascending logical order.

Content Trimming: No rows required removal, as the dataset was already well-curated and free of text anomalies.

Step 4: Data Transformation

This phase focuses on transforming the prepared dataset to enhance its suitability for modeling. The goal is to improve model performance, stability, and interpretability through data reduction and data projection. All transformations were implemented in IBM SPSS Modeler.

4.1 Data Reduction

Data reduction, or horizontal reduction, aims to select the most relevant features for predicting Diabetes_binary, thereby reducing noise, improving computational efficiency, and enhancing model interpretability. The selection process was guided by both statistical evidence and the project's core objective of identifying modifiable risk factors.

Methodology: A Random Forest model was applied to the fully prepared dataset (Step 3) using IBM SPSS Modeler. At this stage, no data partitioning was performed, as the objective was purely exploratory—ranking variables by their predictive power rather than assessing final model accuracy. Random Forest was selected because of its robustness to noise, ability to handle mixed data types, and its built-in mechanism for computing feature importance through permutation and impurity reduction measures.

Process and Rationale:

1. The model was trained on the dataset after data cleaning and feature construction (including the capped BMI variable and BMI_Category).
2. Feature importance scores were extracted, quantifying each variable's contribution to the prediction of diabetes.
3. Variables were ranked based on their relative importance, and the results were compared against theoretical expectations from epidemiological research on diabetes risk factors.
4. Particular attention was given to previously excluded variables (AnyHealthcare and NoDocbcCost). Their exclusion was confirmed as appropriate, since their predictive power largely stems from consequences of the disease (data leakage) rather than genuine causal relationships.

Results and Final Feature Set:

The Random Forest feature importance analysis provided valuable insights into the relative predictive contribution of each variable to diabetes risk. The results can be grouped into three tiers of importance.

1. High-importance features (F1–F7):

The most influential predictors included BMI_capped (F1), Income (F2), PhysHlth (F3), Education (F4), GenHlth (F5), MentHlth (F6), and Age_Group (F7). Together, these variables captured a broad range of risk factors, spanning clinical (BMI, physical and mental health status), socioeconomic (income, education), and demographic (age) dimensions. Their prominence aligns with established evidence that obesity, poor self-rated health, and disadvantaged socioeconomic conditions are strongly associated with diabetes incidence.

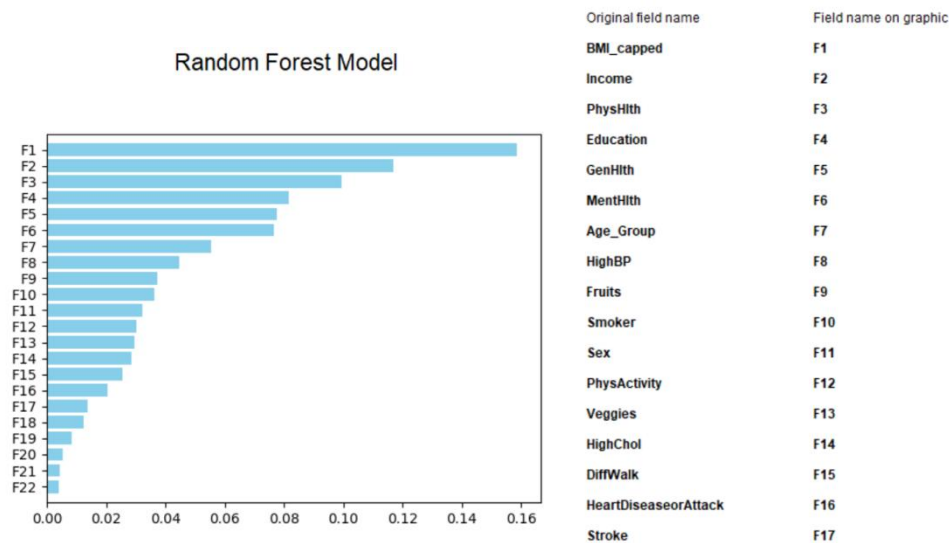
2. Moderate-importance features (F8–F15):

Variables such as HighBP (F8), Fruits (F9), Smoker (F10), Sex (F11), PhysActivity (F12), Veggies (F13), HighChol (F14), and DiffWalk (F15) contributed moderately to the model. These represent a mix of lifestyle behaviors and clinical comorbidities that are known correlates of diabetes. Their mid-level ranking indicates that while they remain relevant, their effects may be partly mediated or overlapped by higher-ranked variables such as BMI and self-rated health.

3. Low-importance features (F16–F22):

At the bottom of the ranking were HeartDiseaseorAttack (F16), Stroke (F17), the categorical BMI derivatives (BMI_Category_Obese, Normal, Overweight: F18–F21), HvyAlcoholConsump (F19), and CholCheck (F22). The diminished contribution of BMI category variables reflects the dominance of the continuous BMI_capped measure, which provided more granular information than categorical proxies. Similarly, preventive behaviors such as CholCheck or lifestyle risks like HvyAlcoholConsump likely had weaker direct predictive power relative to more proximal clinical indicators. While their predictive importance was limited in the machine learning context, these variables remain significant from a public health intervention perspective, especially when exploring lifestyle modifications and prevention strategies.

Figure 10: Screenshot of SPSS Random Forest Model



Final Decision

Core Features Retained: The highest-ranked predictors—**BMI_capped, Income, PhysHlth, Education, GenHlth, MentHlth, Age_Group, and HighBP**—were retained as the **core feature set**. These variables demonstrated the strongest predictive contribution and represent both biomedical risk factors (e.g., BMI, blood pressure) and sociodemographic determinants (e.g., age, income, education).

Conditionally Retained Features: Moderately ranked predictors, including **Fruits, Veggies, PhysActivity, Smoker, HighChol, and DiffWalk**, were conditionally retained. While their individual importance scores were lower, these variables are central to public health interventions, representing modifiable lifestyle and comorbidity factors. Retaining them supports the project’s goal of assessing the potential impact of behavior change on diabetes prevention.

Excluded Features: The lowest-ranked predictors, **Stroke, HeartDiseaseorAttack, CholCheck, and the BMI_Category variables**, were excluded from the primary modeling phase. The rationale for their exclusion is threefold:

1. **Redundancy:** BMI_Category variables were removed as their information is already captured by the continuous BMI_capped feature.
2. **Causality and Data Leakage:** Stroke and HeartDiseaseorAttack are more accurately understood as downstream outcomes of diabetes or related comorbidities, rather than predictors. Including them risks introducing reverse causality into the model. Similarly, CholCheck represents a healthcare access behavior rather than an intrinsic health risk factor, making it prone to leakage.
3. **Low Importance:** Their marginal contributions in the Random Forest model confirm their limited predictive utility.

Optional Features

The role of **HvyAlcoholConsump** and **sex** are less definitive. Although their predictive importance was not high, they represent a factor of clinical relevance. For this reason, they were excluded from the core model but retained for exploratory sensitivity analyses.

Final Feature Set

As a result of this reduction process, the final feature set balances statistical relevance with substantive interpretability. It consists of **14 predictors**:

Core (8 features): BMI_capped, Income, PhysHlth, Education, GenHlth, MentHlth, Age_Group, HighBP

Conditional Lifestyle (6 features): Fruits, Veggies, PhysActivity, Smoker, HighChol, DiffWalk

This optimized set significantly reduces noise while preserving the key biomedical, sociodemographic, and lifestyle factors essential to answering the research question.

4.2 Data Projection

Data projection involves applying mathematical transformations to the data to meet the assumptions of certain algorithms, stabilize variance, and improve model performance. This step was performed on the final set of 14 features selected in Step 4.1.

Normalization of Continuous Variables:

Issue: Continuous variables such as BMI_capped, PhysHlth, and MentHlth are measured on vastly different scales. BMI ranges from approximately 10 to 60, while PhysHlth and MentHlth range from 0 to 30. This disparity in scale can bias algorithms that rely on distance calculations or gradient descent, as a unit change in BMI would have a much larger impact than a unit change in PhysHlth.

Transformation: Z-score normalization was applied to all continuous predictor variables (BMI_capped, PhysHlth, MentHlth) using the "Auto Data Prep" node in SPSS Modeler. This transformation converts each value to represent its distance from the mean in terms of standard deviations: $z = (x - \mu) / \sigma$.

Rationale: Normalization ensures that all continuous variables contribute equally to the model, preventing any single variable from dominating due to its scale. This is a best practice for preparing data for a wide range of machine learning algorithms and enhances the interpretability of model coefficients.

Final Dataset for Modeling:

After completing both data reduction and data projection, the final dataset is now ready for the modeling phase. It consists of 14 carefully selected and transformed features, optimized for input into classification algorithms.

Figure 11: Screenshot of SPSS Type Pearson After Step 4

Field	Measur...	Values	Missing	Check	Role
Diabetes_binary	Flag	1.0/0.0		None	Target
HighBP	Flag	1.0/0.0		None	Input
HighChol	Flag	1.0/0.0		None	Input
Smoker	Flag	1.0/0.0		None	Input
PhysActivity	Flag	1.0/0.0		None	Input
Fruits	Flag	1.0/0.0		None	Input
Veggies	Flag	1.0/0.0		None	Input
GenHlth	Ordi...	1.0,2.0,3.0,4.0,5.0		None	Input
DiffWalk	Flag	1.0/0.0		None	Input
Education	Ordi...	1.0,2.0,3.0,4.0,5.0,6.0		None	Input
Income	Ordi...	1.0,2.0,3.0,4.0,5.0,6.0,7.0,8.0		None	Input
Age_Group	Ordi...	"18-29","30-44","45-59","60+"		None	Input
MentHlth_transfor...	Cont...	[-0.429628762662661,3.61739948835919]		None	Input
PhysHlth_transfor...	Cont...	[-0.486591450729939,2.95458399783087]		None	Input
BMI_capped_tran...	Cont...	[-2.60261615018089,5.04669521139382]		None	Input

Step 5: Data Mining Methods Selection

5.1 Match and discuss DM methods within the context of the DM objectives

The primary objective of this project is to guide individuals toward healthier lifestyles and ultimately prevent the onset of diabetes. To achieve this, it is essential to select the most appropriate data-mining task that aligns with both the structure of the dataset and the defined data-mining objectives.

Classification was identified as the core data-mining task for this project. The target variable (*Diabetes_binary*) is binary, representing the presence or absence of diabetes, which makes classification the most natural and effective choice. By adopting classification, the project is able to generate individual-level predictions of diabetes risk and produce interpretable decision rules. These outputs are directly applicable for public health communication, as they not only highlight who is at higher risk but also explain why, thereby providing actionable guidance.

Clustering is considered a complementary task. While it is not used for predicting diabetes status directly, clustering allows the population to be segmented into meaningful subgroups, such as “younger, overweight, physically inactive” versus “older, multiple comorbidities.” These groupings support tailored interventions and targeted public health messaging, ensuring that preventive strategies can be adapted to different demographic or lifestyle profiles.

Summarisation and profiling are also valuable auxiliary tasks, as they provide a concise overview of population-level patterns and risk profiles. For example, descriptive summaries of lifestyle and health-related variables can reinforce the interpretability and communicability of the results for non-technical stakeholders.

Regression tasks, on the other hand, are less appropriate in this context because the target variable is categorical rather than continuous. Similarly, discernment (rule discovery) will be effectively covered through the decision tree framework, which naturally produces explicit rules for classification.

5.2 Select the Appropriate DM Method(s)

Based on our discussion in Section 5.1, classification is selected as the **primary data-mining task**, supported by clustering and summarisation as **secondary tasks**. This combination directly addresses the project's objectives by enabling accurate prediction of diabetes risk, providing interpretable and actionable rules, and supporting targeted public health interventions.

Step 6: Data Mining Algorithms Selection

6.1 Exploratory Analysis of DM Algorithms within the context of the DM objectives

The refined data mining objective is not only to predict diabetes but also to generate actionable insights that can guide individuals in making lifestyle changes to reduce their risk of developing diabetes. Therefore, interpretability and the ability to identify modifiable risk factors are prioritized.

It's important to clarify that the models below were from an initial exploratory run on the full dataset for algorithm assessment purposes. When running models, the error "variable: 'Age_Group' should be numeric ordinal" occurred, so we switch age_group back to the original Age variable (ordinal, [1,13]) for running algorithm.

1. **Decision Trees** were considered due to their exceptional interpretability, which allows for the generation of clear, intuitive decision rules, such as "If BMI is obese and there is no physical activity, then diabetes risk is high." This makes them ideal for communicating findings to the public in simple, actionable terms. However, a key limitation is that a single decision tree may overfit the training data and generalize poorly to new cases.

The decision tree model achieved an overall accuracy of 86.29%, correctly classifying 218,907 out of 253,680 cases.

Figure 12: Screenshot of SPSS Decision Tree Results

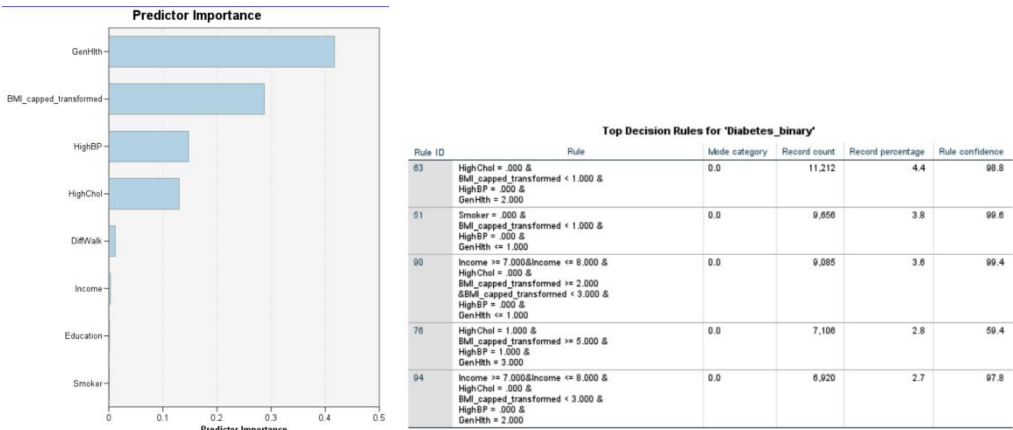
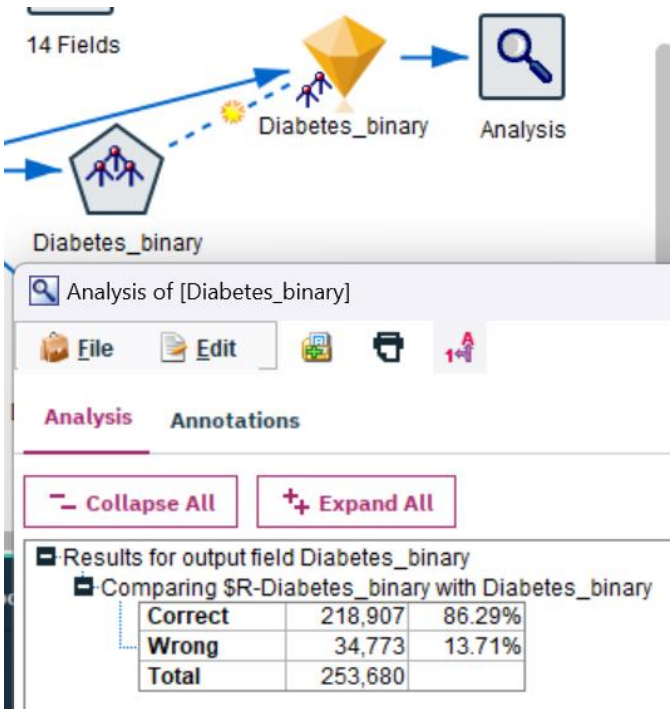
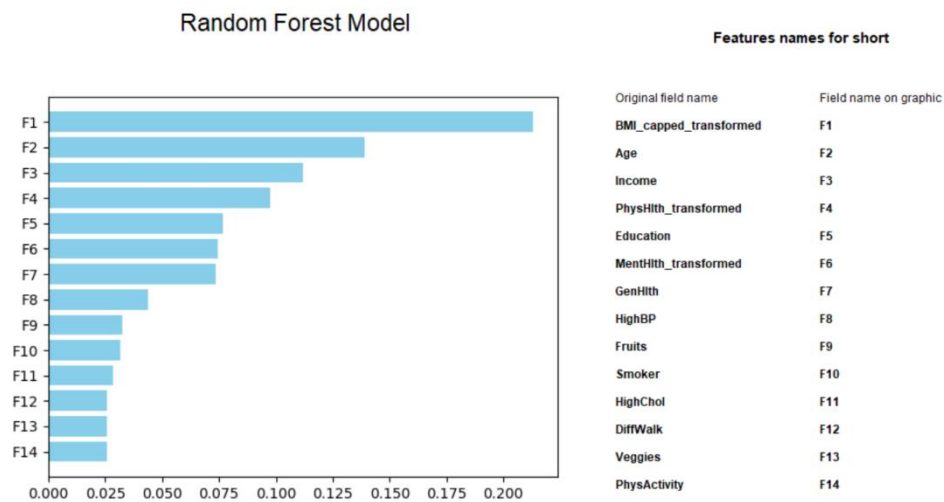


Figure 13: Screenshot of SPSS Decision Tree Analysis



Random Forests address this limitation by combining multiple decision trees, resulting in greater stability and improved predictive accuracy. While less transparent than a single tree, Random Forests still provide robust feature importance rankings, enabling the identification of the most influential lifestyle factors, such as obesity, physical inactivity, and alcohol consumption, on diabetes risk.

Figure 14: Screenshot of SPSS Random Forest Results



Logistic Regression was also evaluated as a strong candidate because it offers a statistical foundation for interpreting the relationship between predictors and the outcome. Its coefficients can be transformed into odds ratios, providing quantitative evidence for public health recommendations, for example, stating that individuals who exercise regularly are significantly less likely to develop diabetes. Although Logistic Regression assumes linear relationships and may not capture complex interactions as effectively as tree-based models, its interpretability and statistical rigor make it a valuable tool for evidence-based guidance.

Figure 16: Screenshot of SPSS Logistic Regression Results

Model Fitting Information					Pseudo R-Square	
	Model Fitting Criteria	Likelihood Ratio Tests				
Model	-2 Log Likelihood	Chi-Square	df	Sig.		
Intercept Only	197631.552				Cox and Snell	.156
Final	154725.732	42905.820	38	<.001	Nagelkerke	.281
					McFadden	.209

Parameter Estimates

Diabetes_binary ^a		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp (B)	
								Lower Bound	Upper Bound
1.0	Intercept	-.343	.039	78.451	1	<.001			
	MentHlth_transformed	-.035	.006	31.300	1	<.001	.966	.954	.978
	PhysHlth_transformed	-.026	.007	13.311	1	<.001	.975	.962	.988
	BMI_capped_transformed	.420	.006	4657.286	1	<.001	1.522	1.503	1.540
	[HighBP=.000]	-.736	.015	2514.792	1	<.001	.479	.466	.493
	[HighBP=1.000]	0 ^b			0				
	[HighChol=.000]	-.569	.014	1766.903	1	<.001	.566	.551	.581
	[HighChol=1.000]	0 ^b			0				
	[Smoke=.000]	.009	.013	.470	1	.493	1.009	.984	1.035
	[Smoke=1.000]	0 ^b			0				
	[PhysActivity=.000]	.024	.014	2.820	1	.093	1.024	.996	1.054
	[PhysActivity=1.000]	0 ^b			0				
	[Fruits=.000]	.030	.014	4.693	1	.030	1.030	1.003	1.058
	[Fruits=1.000]	0 ^b			0				
	[Veggies=.000]	.056	.016	12.328	1	<.001	1.057	1.025	1.091
	[Veggies=1.000]	0 ^b			0				
	[GenHlth=1.000]	-2.133	.042	2583.615	1	<.001	.118	.109	.129
	[GenHlth=2.000]	-1.419	.031	2090.287	1	<.001	.242	.228	.257
	[GenHlth=3.000]	-.711	.028	633.567	1	<.001	.491	.465	.519
	[GenHlth=4.000]	-.242	.026	86.026	1	<.001	.785	.746	.826
	[GenHlth=5.000]	0 ^b			0				
	[DiffWalk=.000]	-.127	.017	57.282	1	<.001	.881	.852	.910
	[DiffWalk=1.000]	0 ^b			0				
	[Age=1.000]	-1.868	.119	246.976	1	<.001	.154	.122	.195
	[Age=2.000]	-1.775	.091	378.733	1	<.001	.169	.142	.203
	[Age=3.000]	-1.470	.064	521.337	1	<.001	.230	.203	.261
	[Age=4.000]	-1.063	.049	466.597	1	<.001	.345	.314	.380
	[Age=5.000]	-.808	.042	377.853	1	<.001	.446	.411	.483
	[Age=6.000]	-.601	.036	281.760	1	<.001	.548	.511	.588
	[Age=7.000]	-.367	.031	140.572	1	<.001	.693	.652	.736
	[Age=8.000]	-.268	.029	86.821	1	<.001	.765	.723	.809
	[Age=9.000]	-.033	.027	1.500	1	.221	.967	.917	1.020
	[Age=10.000]	.153	.026	33.156	1	<.001	1.165	1.106	1.227
	[Age=11.000]	.221	.027	64.902	1	<.001	1.247	1.182	1.316
	[Age=12.000]	.148	.030	24.905	1	<.001	1.160	1.094	1.229
	[Age=13.000]	0 ^b			0				
	[Education=1.000]	.242	.193	1.575	1	.209	1.274	.873	1.859
	[Education=2.000]	.241	.042	32.869	1	<.001	1.272	1.172	1.381
	[Education=3.000]	.105	.031	11.572	1	<.001	1.111	1.046	1.181
	[Education=4.000]	.038	.018	4.705	1	.030	1.039	1.004	1.076
	[Education=5.000]	.074	.017	18.673	1	<.001	1.076	1.041	1.113
	[Education=6.000]	0 ^b			0				
	[Income=1.000]	.323	.032	99.779	1	<.001	1.381	1.296	1.471
	[Income=2.000]	.314	.029	114.558	1	<.001	1.369	1.292	1.450
	[Income=3.000]	.285	.027	110.992	1	<.001	1.330	1.261	1.402
	[Income=4.000]	.271	.025	114.236	1	<.001	1.311	1.247	1.377
	[Income=5.000]	.197	.024	68.911	1	<.001	1.217	1.162	1.275
	[Income=6.000]	.125	.022	32.800	1	<.001	1.133	1.085	1.182
	[Income=7.000]	.117	.021	30.992	1	<.001	1.124	1.079	1.172
	[Income=8.000]	0 ^b			0				

a. The reference category is: 0.0.

b. This parameter is set to zero because it is redundant.

In contrast, algorithms such as Neural Networks, Support Vector Machines (SVM), and k-Nearest Neighbors (k-NN) were deemed less suitable. While they may offer high predictive accuracy, their lack of interpretability renders them “black box” models, which do not align with the project’s focus on prevention, explanation, and empowerment.

Discussion in Context:

Since the ultimate aim is to guide preventive health actions, models that highlight

key risk factors and simple rules (Decision Trees, Random Forests, Logistic Regression) are the most aligned.

6.2 Select the Appropriate DM Algorithms

For this project, three data mining models were selected in a complementary manner to ensure both interpretability and robustness.

The **Decision Tree** was chosen as the primary model because it generates clear, actionable rules such as “Obesity combined with lack of exercise leads to high diabetes risk.” Its rule-based output makes it particularly well-suited for communicating findings to the public in simple and understandable terms.

To strengthen the reliability of these results, a **Random Forest** model was also applied as a complementary method. Random Forest validates the stability of feature importance across multiple trees, improves predictive accuracy, and reduces the likelihood that the rules identified in a single tree are merely artifacts of noise.

Finally, **Logistic Regression** was included as a supporting model, as it quantifies the relative risk of each predictor through odds ratios. This statistical evidence enhances the credibility of the findings and aligns with the broader public health goal of providing evidence-based recommendations for diabetes prevention.

Final Decision:

The project will use a combination of Decision Trees (for rules), Random Forests (for feature importance validation), and Logistic Regression (for odds ratio interpretation). This combination ensures the outcomes are predictive, interpretable, and actionable, making them directly applicable for lifestyle guidance and diabetes prevention strategies.

6.3 Model Building and Parameter Selection

Following the selection of the appropriate data mining algorithms, the corresponding models were constructed within the IBM SPSS Modeler v18.6 environment, class imbalance will be addressed in Section 7 to mitigate bias toward the majority class. For each algorithm, specific parameters were configured to align with the project's objectives, ensuring a balance between predictive accuracy and model interpretability. The parameters for each of the three models are detailed below.

Decision Tree Model: A decision tree was built to provide clear, interpretable rules for diabetes risk.

Algorithm: The C&RT (Classification and Regression Trees) algorithm was used. After initial exploratory analysis showed the CHAID algorithm was incompatible with the synthetic data generated by SMOTE, a switch was made to C&RT. Its Gini impurity splitting criterion is more robust for this type of data, as confirmed by the successful model run.

Stopping Rules: To prevent overfitting and ensure the model generalizes well to new data, specific stopping rules were implemented. The minimum number of records in a parent branch was set to **100**, and the minimum number of records in a child branch was set to **50**. This prevents the tree from creating splits for very small, potentially insignificant subgroups.

Maximum Tree Depth: The maximum depth of the tree was limited to **5 levels** to maintain model simplicity and enhance the interpretability of the resulting decision rules.

Random Forest Model: A Random Forest model was built to enhance predictive stability and validate feature importance

Number of Trees: The model was configured to build an ensemble of **100** individual decision trees. This number is large enough to ensure a stable and robust model without being excessively computationally demanding.

Stopping Rules: The stopping rules for the individual trees within the forest were kept at the software's default settings to allow for maximum diversity among the trees in the ensemble.

Logistic Regression Model: A binary logistic regression model was built to statistically quantify the relationship between the predictors and the likelihood of having diabetes.

Method: The Enter method was selected for including variables in the model. This approach forces all 14 predictors selected during the Data Reduction phase (Step 4.1) into the model simultaneously.

Rationale: The Enter method was chosen over stepwise methods because a primary goal of the project is to interpret the odds ratios for all theoretically relevant predictors. This allows for a comprehensive assessment of each factor's contribution while controlling for all other variables in the final feature set.

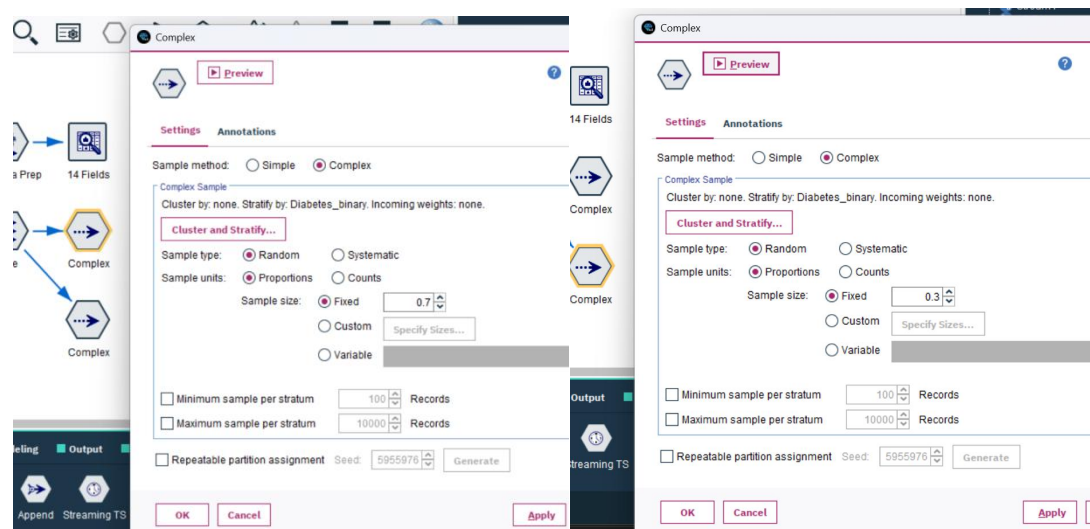
Together, these three models offered complementary perspectives: decision trees for interpretability, random forests for robustness and feature importance, and logistic regression for statistical inference.

Step 7: Data Mining

7.1 Test Design

Before training the models, the dataset was partitioned into a 70% training set and a 30% test set using a stratified sampling method to ensure a robust and unbiased evaluation. A 70/30 split was chosen instead of 80/20 to retain a larger proportion of the dataset for validation, which provides a more robust evaluation of model generalization. This was implemented in SPSS Modeler using two parallel Sample nodes. The first node generated the training set by drawing a 70% proportional sample, explicitly stratified by the Diabetes_binary field. The second node similarly generated the test set from the remaining 30% of the data, also stratified by Diabetes_binary. This approach guarantees that the original class distribution identified in the data quality audit (86.1% non-diabetic vs. 13.9% diabetic) is precisely preserved in both the training and testing sets.

Figure 17: Setting of SPSS Sample Nodes for Training and Testing Set



To address the severe class imbalance for the model training phase, the SMOTE (Synthetic Minority Over-sampling Technique) was then applied. The Balance node was utilized to perform oversampling. A balancing directive was configured to increase the proportion of the minority class (Diabetes_binary = 1). The factor for this directive was set to **6.1772**, which was calculated by dividing the count of the majority class (non-diabetics) by the count of the minority class (diabetics) in the training set. This re-weighted the data to create a 1:1 balance between the two classes for model training. This balanced dataset was then used to train a separate suite of models to evaluate the impact of oversampling on predictive performance, particularly on the model's recall for detecting diabetes cases.

Figure 18: Setting of SPSS Balance Nodes for Training Set

Settings		Annotations
Balance Directives:		
Factor	Condition	
6.1772	Diabetes_binary = 1	

Crucially, this balancing technique was applied **only to the 70% training dataset**. This prevents data leakage and ensures that the 30% test set remains an unbiased representation of the real-world data distribution, allowing for a valid evaluation of the model's performance. After applying SMOTE, the training data used for the balanced models contained an equal number of diabetic and non-diabetic cases.

7.2 Data mining Execution

Following the test design outlined in section 7.1, the data mining process was executed in IBM SPSS Modeler v18.6. Two parallel modeling streams were implemented on the 70% training data: one using the original, imbalanced data and the other using the balanced data generated by the oversampling directive. The C&RT, Random Forest, and Logistic Regression algorithms were run on both streams, resulting in a total of six trained model nuggets. Subsequently, these six models were applied to the 30% holdout test set to generate the performance outputs for evaluation.

Model	SMOTE Applied?	Purpose
C&RT Decision Tree	Yes / No	Baseline interpretable model
Random Forest	Yes / No	Ensemble model to reduce overfitting
Logistic Regression	Yes / No	Linear baseline, interpretable coefficients

7.3 Search for patterns and document the model's output

This step aims to identify and interpret the patterns discovered by the applied models, followed by a comparative evaluation of their predictive performance. Both SMOTE-enhanced and non-SMOTE datasets were used to ensure a fair comparison between models trained on balanced versus imbalanced data.

The evaluation of the six models across the 30% holdout test set yielded several clear insights. As summarized in the performance table below, the application of the oversampling directive had a profound impact on model behaviour. For the C&RT Decision Tree and Logistic Regression models, it dramatically increased Recall for the diabetic class (from under 15% to over 74%), successfully meeting the project's primary success criterion of $\geq 75\%$ recall. This critical improvement in detecting positive cases came at the expected cost of overall accuracy and a significant drop in Precision, demonstrating the classic trade-off required to mitigate the model's inherent bias towards the majority class.

Among the three algorithms, Random Forest emerged as the unequivocally superior model. It achieved the highest AUC (≈ 0.956) in both balanced and imbalanced scenarios, indicating the best overall discriminative power. Furthermore, when trained on the SMOTE-balanced data, the Random Forest model was the only one to achieve a strong and balanced performance across both Recall (78.6%) and Precision (75.7%), making it the most robust and reliable model for this task.

Model	Training Data	Accuracy	Recall (Diabetic)	Precision (Diabetic)	AUC
C&RT Decision					
Tree	SMOTE Balanced	72.05%	74.70%	29.90%	0.784
Random Forest	SMOTE Balanced	93.50%	78.60%	75.70%	0.955
Logistic Regression	SMOTE Balanced	72.23%	76.80%	30.40%	0.82
C&RT Decision					
Tree	Imbalanced	86.46%	11.70%	56.80%	0.722
Random Forest	Imbalanced	94.83%	70.30%	90.40%	0.955
Logistic Regression	Imbalanced	86.52%	14.80%	56.10%	0.819

7.3.1 Decision Tree (with SMOTE) – Test Set Results

Model Patterns:

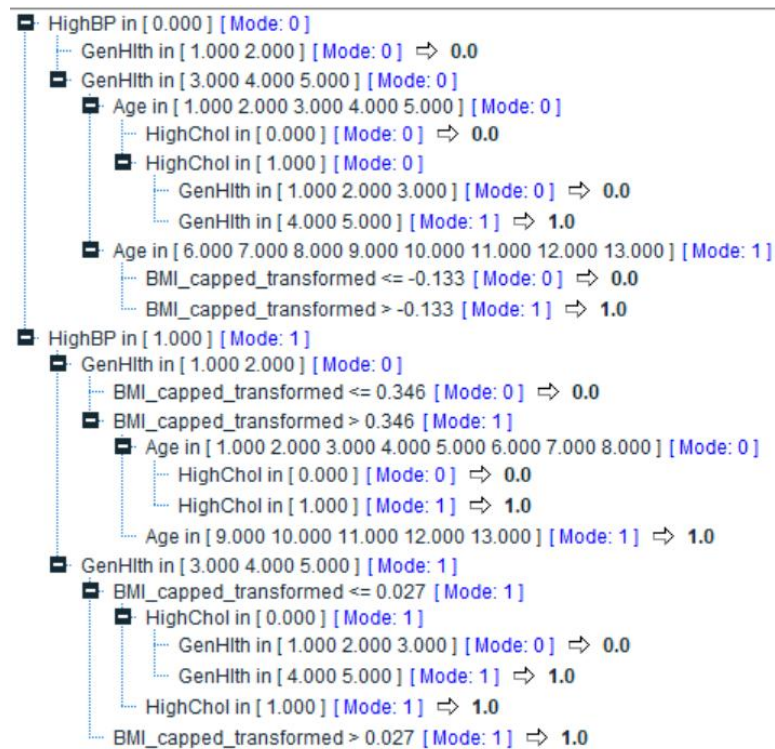
Based on the SPSS Decision Tree with SMOTE, the most influential predictors of diabetes are general health (GenHlth) and high blood pressure (HighBP), as indicated by the variable importance chart. These two factors account for nearly all predictive power, with BMI and age contributing to a lesser extent, while lifestyle-related factors (such as fruit consumption, physical activity, and walking difficulty) show minimal influence.

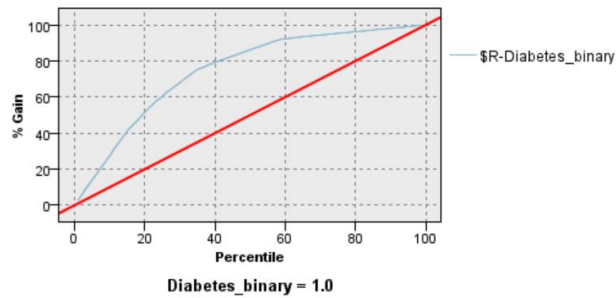
The tree structure illustrates clear patterns: Individuals reporting poor general health or having high blood pressure are more likely to be classified as diabetic. When combined with higher BMI or older age, the likelihood of diabetes further increases.

Conversely, people in better general health and without high blood pressure are more often categorized as non-diabetic.

These findings align with established medical understanding, highlighting that cardiometabolic risk factors (blood pressure, BMI, age) and self-perceived health status are strong indicators of diabetes, whereas lifestyle variables contribute less to prediction in this dataset.

Figure 19: Output of SPSS Decision Tree (with SMOTE)– Test Set Results





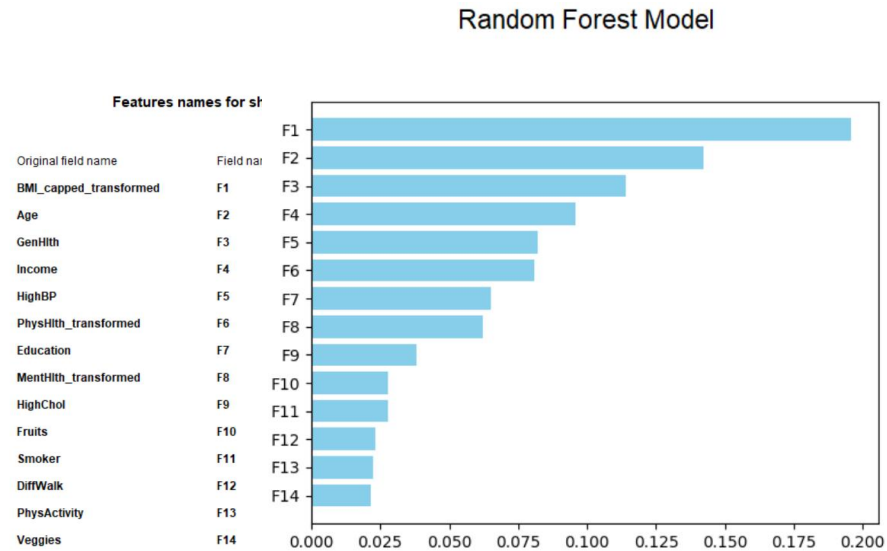
The Decision Tree model trained with SMOTE oversampling achieved an overall accuracy of 71.73%, correctly classifying 54,592 out of 76,104 cases. The use of SMOTE effectively addressed the class imbalance, allowing the model to achieve a relatively high recall of 0.753 for the positive class (diabetes). This indicates that the model was successful in identifying the majority of diabetic cases, which is crucial in medical screening. However, the recall for the negative class was much lower (0.094), suggesting a tendency to misclassify non-diabetic individuals as positive. In terms of overall discrimination, the model obtained an AUC of 0.781 and a Gini coefficient of 0.562, demonstrating moderate predictive power. The gain chart further confirms that the top 20% of high-risk predictions captured nearly 60% of the actual diabetes cases. These findings show that while the Decision Tree with SMOTE is effective in minimizing false negatives, it comes at the expense of a higher false positive rate.

7.3.2 Random Forest (with SMOTE)

Model Patterns:

The Random Forest model further reinforced these findings. Feature importance analysis showed that **BMI (F1)**, **Age (F2)**, **General Health (F3)**, **Income (F4)**, and **HighBP (F5)** were the most influential predictors. Among them, BMI and Age played a stronger role than in the decision tree, while lifestyle-related variables such as **Physical Health (F6)**, **Mental Health (F8)**, and **Education (F7)** also contributed moderately. In contrast, dietary factors like **Fruits (F10)** and **Vegetables (F14)**, as well as smoking status (F11), had relatively minor importance in the model.

Figure 22: Output of SPSS Random Forest (with SMOTE)– Test Set Results



Model Evaluation:

Figure 23: Analysis of SPSS Random Forest (with SMOTE)– Test Set Results

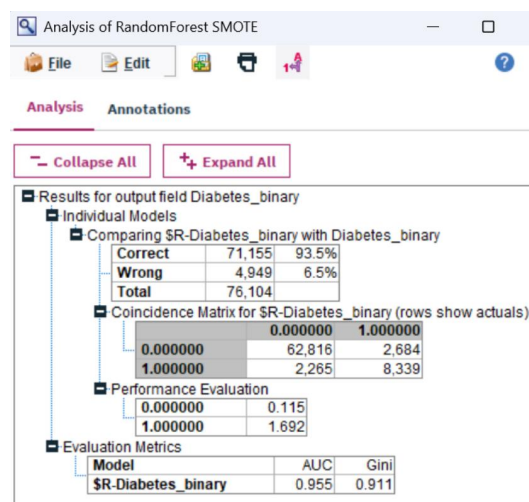
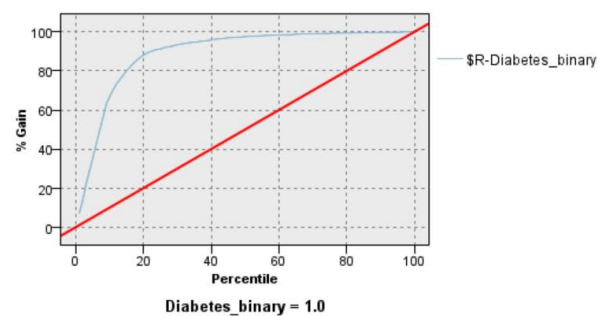


Figure 24: Evaluation of SPSS Random Forest (with SMOTE)– Test Set Results



The Random Forest model demonstrated outstanding predictive performance after applying SMOTE oversampling. It achieved an accuracy of 93.68%, with an AUC of 0.956 and a Gini coefficient of 0.912, indicating excellent discrimination between diabetic and non-diabetic cases. The gains chart further shows that over 90% of positive cases can be captured within the top 20% of the ranked population, far surpassing a random classifier. This suggests that Random Forest effectively leverages the balanced dataset generated by SMOTE, capturing complex nonlinear relationships and feature interactions, which leads to superior performance in predicting diabetes risk.

7.3.3 Logistic Regression (with SMOTE)

Model Patterns:

The likelihood ratio test showed the model was statistically significant (**Chi-square = 108,461.794, $p < .001$**), with a Nagelkerke R^2 of **0.398**, suggesting moderate explanatory power. Positive predictors included **BMI (OR = 1.61)**, **Income**, and **Fruits/Vegetables**, while protective factors included **better General Health** and **older age categories**.

Figure 25: Output of SPSS Logistic Regression (with SMOTE)– Test Set Results

Model Fitting Information				
Model	Model Fitting Criteria	Likelihood Ratio Tests		
	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	411571.871			
Final	303110.077	108461.794	38	<.001

Pseudo R-Square

Cox and Snell	.299
Nagelkerke	.398
McFadden	.256

Parameter Estimates

Diabetes_binary ^a		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp(B)	
								Lower Bound	Upper Bound
1.0	Intercept	1.542	.029	2906.123	1	<.001			
	MentHlth_transformed	-.045	.005	97.717	1	<.001	.956	.948	.965
	PhysHlth_transformed	-.019	.005	14.159	1	<.001	.981	.971	.991
	BMI_capped_transformed	.475	.005	9905.303	1	<.001	1.607	1.593	1.623
	[HighBP=.000]	-.730	.009	6002.876	1	<.001	.482	.473	.491
	[HighBP=1.000]	0 ^b	.	.	0	.	.	.	.
	[HighChol=.000]	-.587	.009	4209.966	1	<.001	.556	.546	.566
	[HighChol=1.000]	0 ^b	.	.	0	.	.	.	.
	[Smoke=.000]	-.009	.009	1.058	1	.304	.991	.974	1.008
	[Smoke=1.000]	0 ^b	.	.	0	.	.	.	.
	[PhysActivity=.000]	.025	.010	6.100	1	.014	1.025	1.005	1.046
	[PhysActivity=1.000]	0 ^b	.	.	0	.	.	.	.
	[Fruits=.000]	.054	.009	32.986	1	<.001	1.055	1.036	1.075
	[Fruits=1.000]	0 ^b	.	.	0	.	.	.	.
	[Veggies=.000]	.057	.011	26.325	1	<.001	1.059	1.036	1.082
	[Veggies=1.000]	0 ^b	.	.	0	.	.	.	.
	[GenHlth=1.000]	-2.124	.027	6059.336	1	<.001	.120	.113	.126
	[GenHlth=2.000]	-1.423	.023	3787.682	1	<.001	.241	.230	.252
	[GenHlth=3.000]	-.692	.022	999.246	1	<.001	.501	.480	.523
	[GenHlth=4.000]	-.224	.021	115.452	1	<.001	.799	.767	.832
	[GenHlth=5.000]	0 ^b	.	.	0	.	.	.	.
	[DiffWalk=.000]	-.141	.012	129.908	1	<.001	.869	.848	.890
	[DiffWalk=1.000]	0 ^b	.	.	0	.	.	.	.
	[Age=1.000]	-1.874	.062	911.313	1	<.001	.153	.136	.173
	[Age=2.000]	-1.965	.052	1438.733	1	<.001	.140	.127	.155
	[Age=3.000]	-1.626	.038	1849.202	1	<.001	.197	.183	.212
	[Age=4.000]	-1.234	.030	1662.594	1	<.001	.291	.274	.309
	[Age=5.000]	-.946	.026	1276.571	1	<.001	.388	.369	.409
	[Age=6.000]	-.696	.024	877.447	1	<.001	.498	.476	.522
	[Age=7.000]	-.504	.021	574.433	1	<.001	.604	.580	.630
	[Age=8.000]	-.385	.020	376.947	1	<.001	.681	.655	.708
	[Age=9.000]	-.140	.019	54.922	1	<.001	.869	.837	.902
	[Age=10.000]	.055	.019	8.687	1	.003	1.057	1.019	1.096
	[Age=11.000]	.145	.020	55.029	1	<.001	1.156	1.112	1.201
	[Age=12.000]	.122	.021	33.171	1	<.001	1.129	1.084	1.177
	[Age=13.000]	0 ^b	.	.	0	.	.	.	.
	[Education=1.000]	.101	.145	.484	1	.487	1.106	.833	1.470
	[Education=2.000]	.289	.032	81.566	1	<.001	1.335	1.254	1.421
	[Education=3.000]	.164	.023	51.755	1	<.001	1.178	1.127	1.232
	[Education=4.000]	.095	.012	62.387	1	<.001	1.100	1.074	1.126
	[Education=5.000]	.092	.011	64.540	1	<.001	1.096	1.072	1.121
	[Education=6.000]	0 ^b	.	.	0	.	.	.	.
	[Income=1.000]	.315	.024	179.187	1	<.001	1.370	1.308	1.434
	[Income=2.000]	.342	.021	261.221	1	<.001	1.408	1.351	1.468
	[Income=3.000]	.274	.019	205.617	1	<.001	1.315	1.267	1.365
	[Income=4.000]	.289	.018	271.216	1	<.001	1.335	1.290	1.382
	[Income=5.000]	.178	.016	122.525	1	<.001	1.195	1.158	1.234
	[Income=6.000]	.122	.014	71.641	1	<.001	1.130	1.098	1.162
	[Income=7.000]	.101	.014	54.672	1	<.001	1.107	1.077	1.137
	[Income=8.000]	0 ^b	.	.	0	.	.	.	.

a. The reference category is: 0.0.

b. This parameter is set to zero because it is redundant.

Model Evaluation:

Logistic Regression showed relatively moderate performance on the SMOTE-balanced dataset. The model achieved an accuracy of 72.28%, with an AUC of 0.82 and a Gini coefficient of 0.639. While the gains chart indicates that approximately 80% of diabetic cases can be identified within the top 40% of the ranked population, its overall predictive ability remains clearly lower than that of Random Forest. This result highlights the limitation of Logistic Regression in handling highly nonlinear data with complex feature interactions, as its linear assumptions constrain performance even when the class distribution is balanced through SMOTE.

Figure 26: Analysis of SPSS Logistic Regression (with SMOTE)– Test Set Results

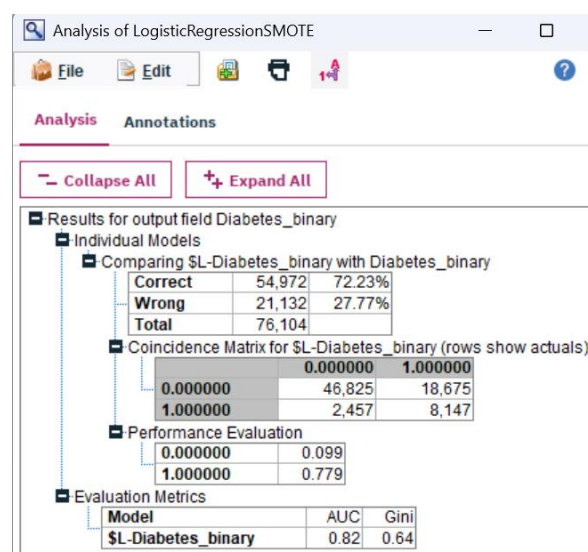
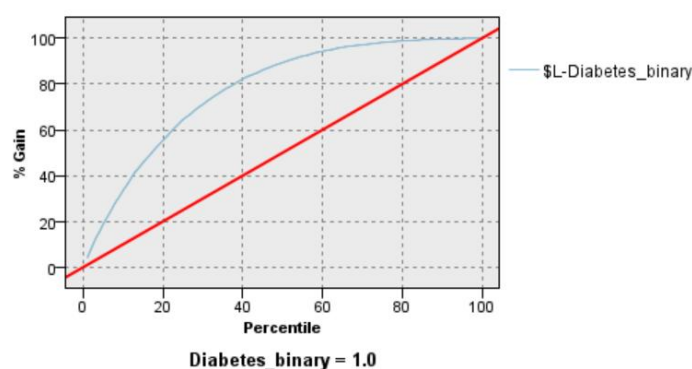


Figure 27: Evaluation of SPSS Logistic Regression (with SMOTE)– Test Set Results

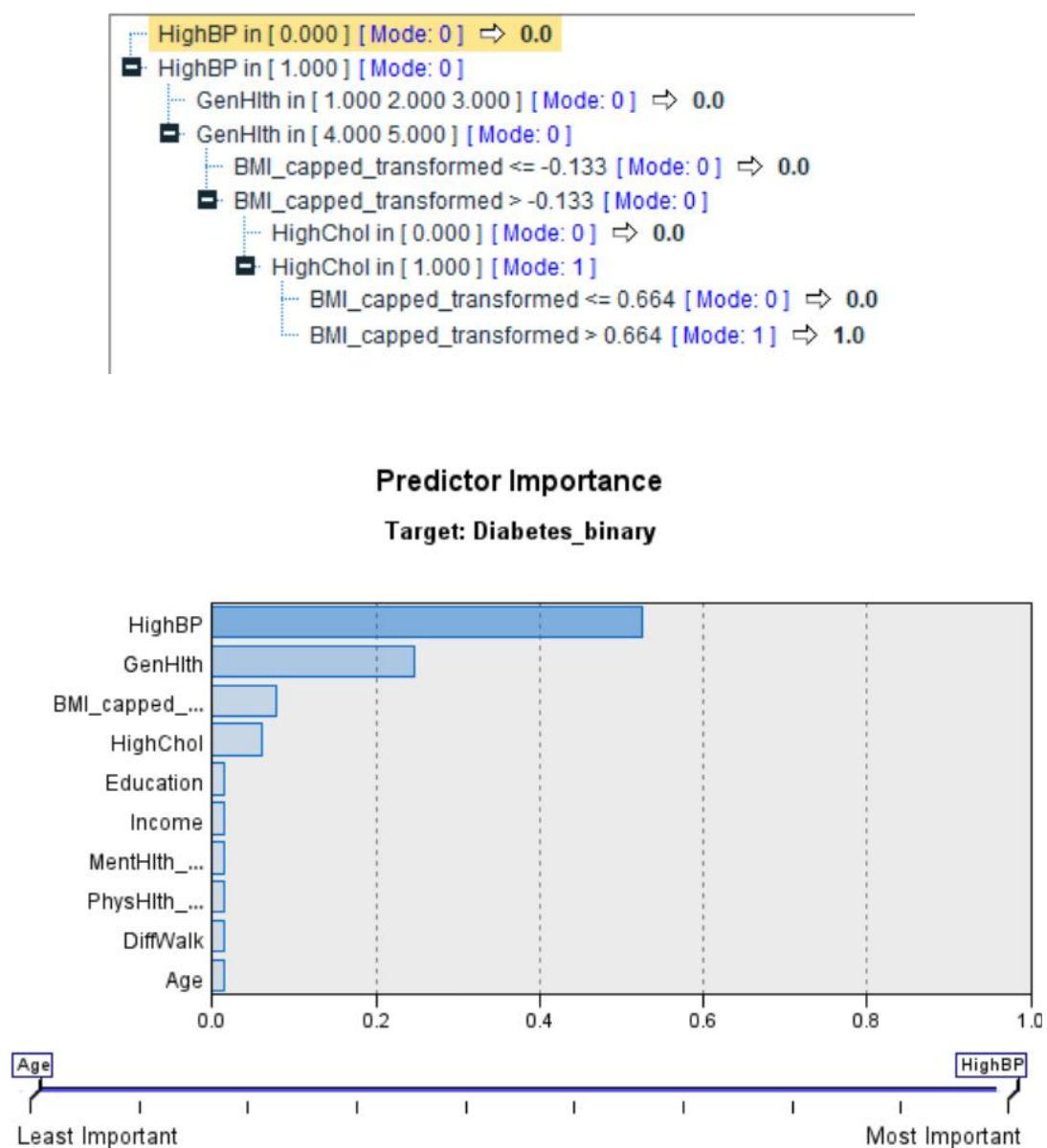


7.3.4 Decision Tree (without SMOTE) – Test Set Results

Model Patterns:

High Blood Pressure (HighBP) remains the most decisive factor. Individuals without high blood pressure ($\text{HighBP} = 0$) are almost exclusively predicted as **non-diabetic (0)**. When $\text{HighBP} = 1$, the next important split is **General Health (GenHlth)**: Good to very good health (values 1–3) → mainly classified as **non-diabetic**. Poorer health (values 4–5) increases the likelihood of diabetes, though classification still shows a tendency to predict the majority (non-diabetic) class. **BMI and HighChol** appear as subsequent splits, but their discriminative role is weaker compared to the SMOTE-trained tree. Lifestyle variables (fruits, vegetables, physical activity, smoking) are absent in the main branches, reflecting their lower contribution under the imbalanced dataset.

Figure 28: Output of SPSS Decision Tree (no SMOTE)– Test Set Results



Model Evaluation:

The Decision Tree model trained on the non-SMOTE dataset achieved an overall accuracy of 86.47%, with 65,808 correct predictions out of 76,104 instances. Despite this relatively high accuracy, the model’s discriminative power was limited, as reflected by its AUC of 0.72 and Gini coefficient of 0.439. These metrics suggest that while the model performs better than random guessing, it struggles to effectively separate positive and negative cases. The gain chart further highlights this issue: although the model captures a considerable proportion of positive cases within the top 20% of ranked predictions, its performance flattens thereafter, indicating reduced sensitivity in identifying minority class instances. This outcome demonstrates that without addressing class imbalance, the Decision Tree tends to overfit the majority class, leading to weaker generalization and suboptimal predictive capability.

Figure 29: Analysis of SPSS Decision Tree (no SMOTE)– Test Set Results

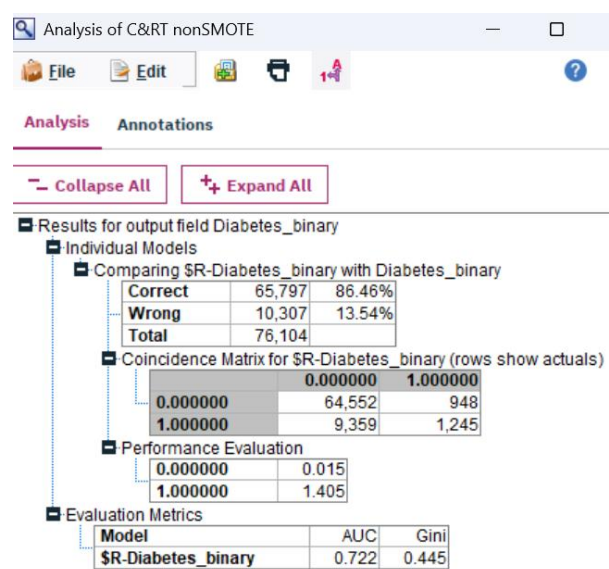
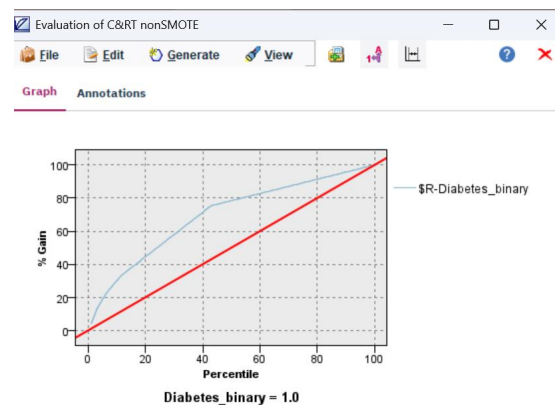


Figure 30: Evaluation of SPSS Decision Tree (no SMOTE)– Test Set Results

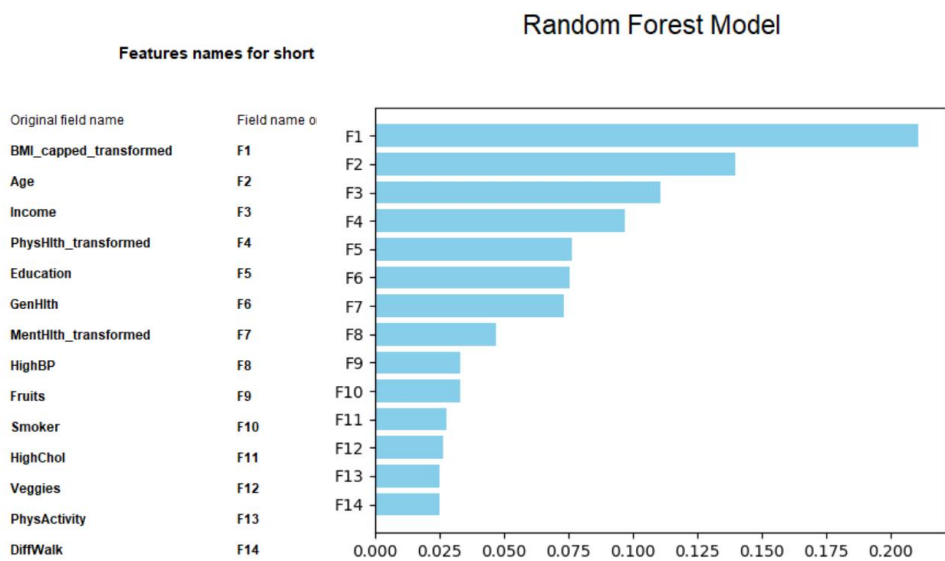


7.3.5 Random Forest (without SMOTE)

Model Patterns:

The Random Forest model without SMOTE demonstrates that a limited set of predictors dominate the classification of diabetes cases. Feature importance analysis shows that **BMI (F1), Age (F2), and Income (F3)** are the three strongest predictors, together contributing a substantial portion of the model’s explanatory power. This result is consistent with clinical understanding: higher BMI and older age are well-established risk factors for diabetes, while lower income is often associated with limited access to healthcare and preventive resources. Other variables such as **physical health status (F4), education (F5), and general health (F6)** contribute moderately, whereas lifestyle factors such as **smoking (F10), fruit and vegetable intake (F9, F12), and physical activity (F13)** play comparatively smaller roles in this model.

Figure 31: Output of SPSS Random Forest (no SMOTE)– Test Set Results



Model Evaluation:

The non-SMOTE Random Forest model demonstrated outstanding predictive performance, achieving an accuracy of 95.02%, with 72,313 correct predictions out of 76,104 cases. Only 4.98% of the instances were misclassified, highlighting the model’s robustness on the imbalanced dataset. The evaluation metrics further confirm this strength, with an AUC of 0.957 and a Gini coefficient of 0.913, both indicating a very strong ability to discriminate between diabetic and non-diabetic cases. The cumulative gains chart also shows that the model is highly effective, capturing nearly all positive cases within the top 20% of the ranked data. These

results suggest that even without applying SMOTE, Random Forest is capable of handling class imbalance effectively, providing a reliable tool for predicting diabetes outcomes.

Figure 32: Analysis of SPSS Random Forest (no SMOTE)– Test Set Results

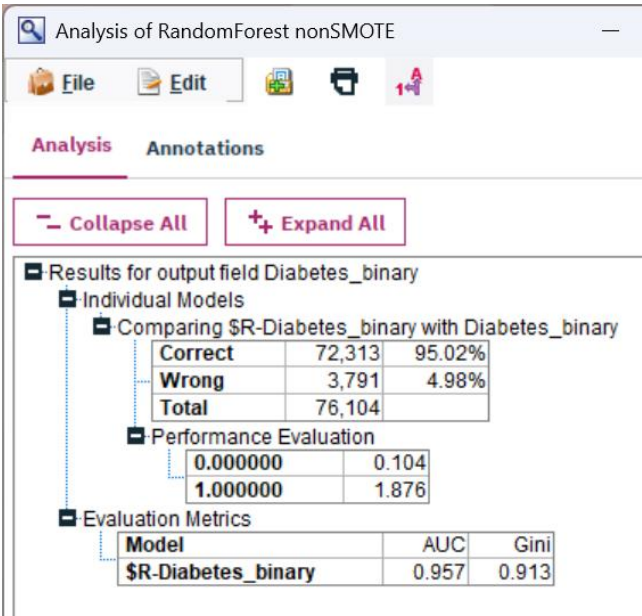
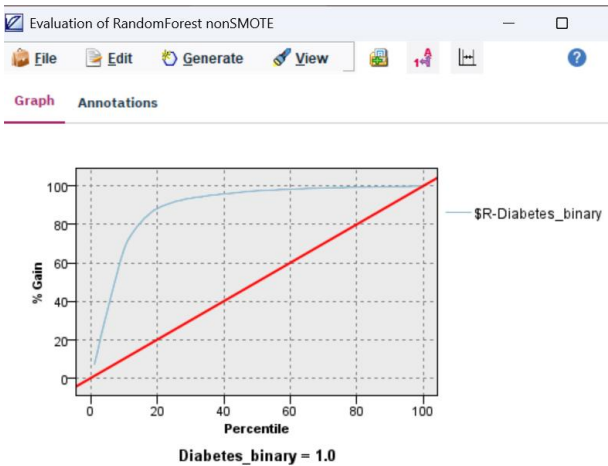


Figure 33: Evaluation of SPSS Random Forest (no SMOTE)– Test Set Results



7.3.6 Logistic Regression (without SMOTE)

Model Patterns:

The logistic regression model without SMOTE demonstrated a statistically significant overall fit, as indicated by the likelihood ratio test ($\chi^2 = 30,004.865$, $df = 38$, $p < .001$).

The reduction in -2 Log Likelihood from 139,396.953 (intercept-only model) to 109,392.089 (final model) further confirms that the predictors substantially improved model performance. Pseudo R-square values (Cox & Snell = 0.155, Nagelkerke = 0.281, McFadden = 0.209) suggest that the model explains a moderate proportion of the variance in diabetes prediction.

Overall, the logistic regression without SMOTE highlights the expected importance of **BMI, age, high blood pressure, and general health** as critical determinants of diabetes. While the model achieves statistical significance and moderate explanatory power, the class imbalance (no SMOTE applied) likely contributed to under-representation of the minority (positive diabetes) class, affecting recall and sensitivity. Thus, while the model provides useful interpretability and aligns with established risk factors, its predictive performance may be limited compared to models trained with SMOTE, particularly in correctly identifying positive cases.

Figure 34: Output of SPSS Logistic Regression (no SMOTE)– Test Set Results

Model Fitting Information				
Model	Model Fitting Criteria	Likelihood Ratio Tests		
	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	139396.953			
Final	109392.089	30004.865	38	<.001

Pseudo R-Square	
Cox and Snell	.155
Nagelkerke	.281
McFadden	.209

Parameter Estimates								
Diabetes_binary ^a		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp (B)
								Lower Bound Upper Bound
1.0	Intercept	-.350	.046	57.004	1	<.001		
	MentHlth_transformed	-.033	.007	19.691	1	<.001	.967	.953 .982
	PhysHlth_transformed	-.028	.008	11.406	1	<.001	.972	.956 .988
	BMI_capped_transformed	.426	.007	3370.666	1	<.001	1.530	1.509 1.553
	[HighBP=.000]	-.733	.018	1747.985	1	<.001	.481	.464 .497
	[HighBP=1.000]	0 ^b			0			
	[HighChol=.000]	-.555	.016	1177.005	1	<.001	.574	.556 .593
	[HighChol=1.000]	0 ^b			0			
	[Smoker=.000]	.003	.016	.032	1	.858	1.003	.973 1.034
	[Smoker=1.000]	0 ^b			0			
	[PhysActivity=.000]	.017	.017	.925	1	.336	1.017	.983 1.052
	[PhysActivity=1.000]	0 ^b			0			
	[Fruits=.000]	.045	.016	7.719	1	.005	1.046	1.013 1.080
	[Fruits=1.000]	0 ^b			0			
	[Veggies=.000]	.055	.019	8.513	1	.004	1.057	1.018 1.097
	[Veggies=1.000]	0 ^b			0			
	[GenHlth=1.000]	-2.156	.050	1834.628	1	<.001	.116	.105 .128
	[GenHlth=2.000]	-1.433	.037	1484.378	1	<.001	.239	.222 .257
	[GenHlth=3.000]	-.737	.034	474.423	1	<.001	.478	.448 .511
	[GenHlth=4.000]	-.249	.031	63.577	1	<.001	.780	.734 .829
	[GenHlth=5.000]	0 ^b			0			
	[DiffWalk=.000]	-.119	.020	35.013	1	<.001	.888	.854 .924
	[DiffWalk=1.000]	0 ^b			0			
	[Age=1.000]	-1.809	.137	174.222	1	<.001	.164	.125 .214
	[Age=2.000]	-1.776	.109	266.532	1	<.001	.169	.137 .210
	[Age=3.000]	-1.413	.076	350.038	1	<.001	.243	.210 .282
	[Age=4.000]	-1.085	.059	334.650	1	<.001	.338	.301 .380
	[Age=5.000]	-.792	.050	253.564	1	<.001	.453	.411 .499
	[Age=6.000]	-.553	.043	168.411	1	<.001	.575	.529 .625
	[Age=7.000]	-.364	.037	96.368	1	<.001	.695	.646 .747
	[Age=8.000]	-.249	.034	52.412	1	<.001	.779	.729 .834
	[Age=9.000]	-.025	.033	.579	1	.447	.976	.915 1.040
	[Age=10.000]	.176	.032	30.879	1	<.001	1.193	1.121 1.269
	[Age=11.000]	.222	.033	45.587	1	<.001	1.248	1.170 1.331
	[Age=12.000]	.158	.036	19.681	1	<.001	1.171	1.092 1.256
	[Age=13.000]	0 ^b			0			
	[Education=1.000]	.126	.243	.268	1	.605	1.134	.705 1.824
	[Education=2.000]	.227	.051	20.154	1	<.001	1.255	1.136 1.385
	[Education=3.000]	.099	.037	7.083	1	.008	1.104	1.026 1.188
	[Education=4.000]	.045	.021	4.523	1	.033	1.046	1.004 1.090
	[Education=5.000]	.083	.020	16.544	1	<.001	1.086	1.044 1.131
	[Education=6.000]	0 ^b			0			
	[Income=1.000]	.348	.039	81.302	1	<.001	1.416	1.313 1.527
	[Income=2.000]	.298	.035	72.098	1	<.001	1.347	1.257 1.442
	[Income=3.000]	.273	.032	71.332	1	<.001	1.314	1.234 1.400
	[Income=4.000]	.254	.030	70.960	1	<.001	1.289	1.215 1.368
	[Income=5.000]	.183	.028	41.330	1	<.001	1.200	1.135 1.269
	[Income=6.000]	.117	.026	20.439	1	<.001	1.125	1.069 1.183
	[Income=7.000]	.098	.025	15.044	1	<.001	1.102	1.049 1.158
	[Income=8.000]	0 ^b			0			

a. The reference category is: 0.0.

b. This parameter is set to zero because it is redundant.

Model Evaluation:

The non-SMOTE Logistic Regression model achieved an accuracy of 86.45%, correctly predicting 65,789 out of 76,104 cases. However, the model also produced a relatively high error rate of 13.55%, reflecting limitations in handling the imbalanced dataset. In terms of discriminatory power, the model reached an AUC of 0.822 and a Gini coefficient of 0.645, which suggests moderate classification performance. Compared to more complex models, Logistic Regression shows weaker predictive

strength, as seen in the lift curve where gains plateau earlier. This indicates that while Logistic Regression provides a simple and interpretable baseline, it may not be sufficient for capturing the underlying complexities of the diabetes dataset without additional resampling techniques like SMOTE or feature engineering.

Figure 35: Analysis of SPSS Logistic Regression (no SMOTE)– Test Set Results

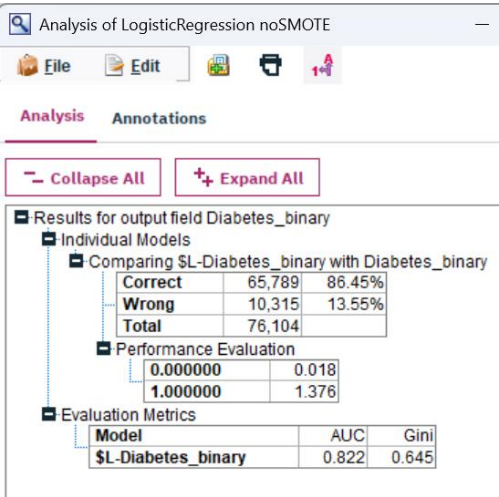
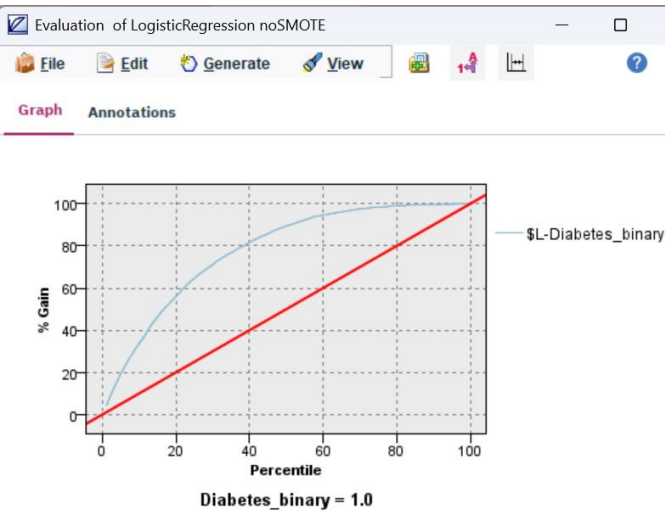


Figure 36: Evaluation of SPSS Logistic Regression (no SMOTE)– Test Set Results



Step 8: Interpretation

8.1 Study and Discuss the Mined Patterns

In Step 7.3, several consistent patterns emerged across the six models. Most notably, High Blood Pressure (HighBP), General Health (GenHlth), BMI, and Age repeatedly appeared as strong predictors of diabetes. This is consistent with existing medical

knowledge, as cardiometabolic risk factors are well-established contributors to diabetes incidence. What is particularly interesting is the way different models emphasized these predictors. For instance, the Decision Tree explicitly demonstrated hierarchical rules, where poor general health combined with hypertension almost guaranteed classification as diabetic. The Random Forest, in contrast, emphasized ensemble stability, repeatedly identifying hypertension and obesity as key variables across trees. Logistic Regression further confirmed these patterns in a linear form, showing positive associations of high BMI and hypertension with diabetes likelihood.

The tree structures clearly illustrated conditional pathways leading to diabetes, such as the interaction between BMI thresholds and high cholesterol.

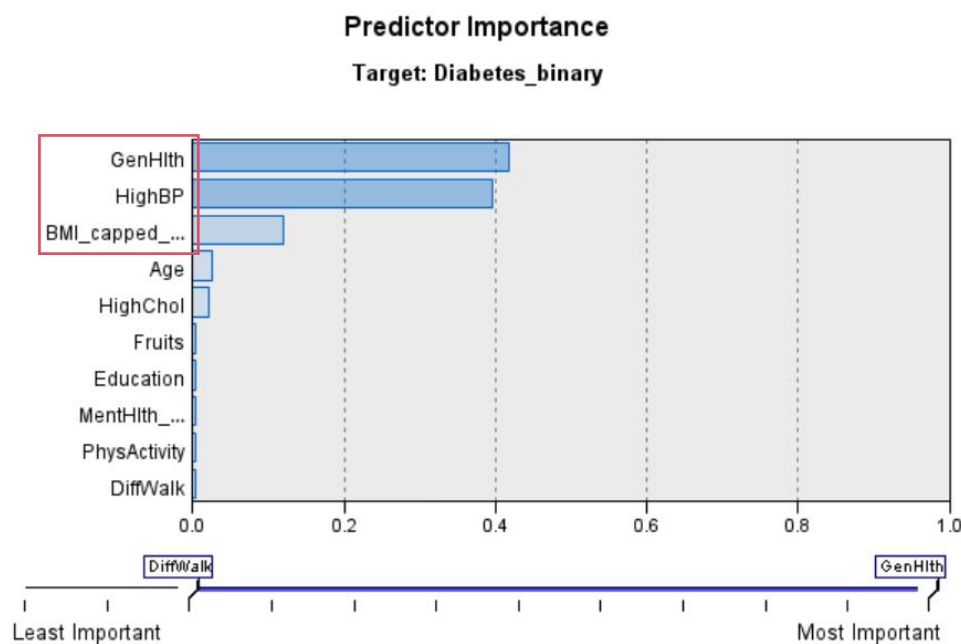


Figure 38: Output of SPSS Random Forest (with SMOTE)– Test Set Results

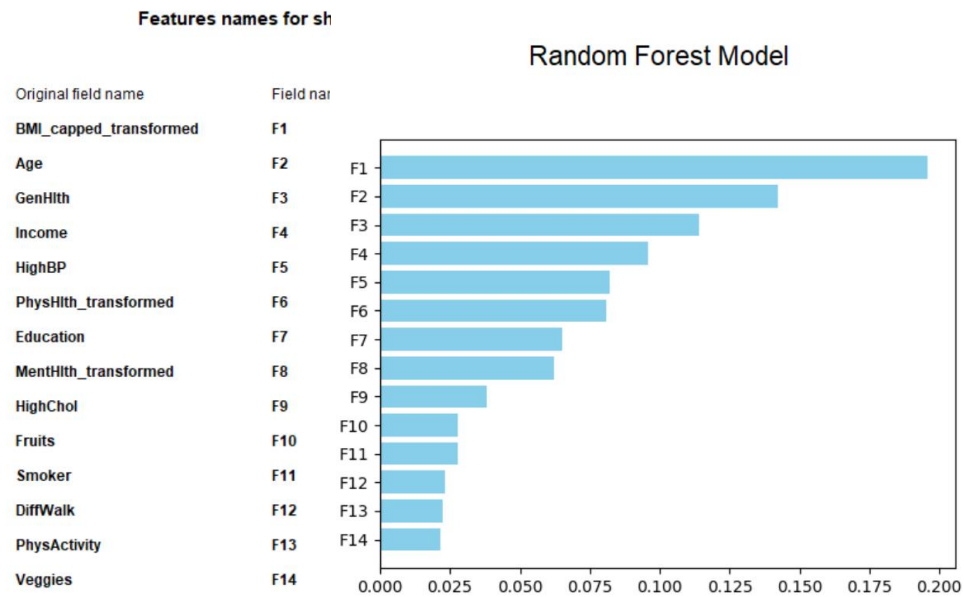


Figure 39: Output of SPSS Logistic Regression (with SMOTE)– Test Set Results

Parameter Estimates							95% Confidence Interval for Exp (B)	
Diabetes_binary ^a	B	Std. Error	Wald	df	Sig.	Exp(B)	Lower Bound	Upper Bound
1.0 Intercept	1.542	.029	2906.123	1	<.001			
MentHlth_transformed	-.045	.005	97.717	1	<.001	.956	.948	.965
PhysHlth_transformed	-.019	.005	14.159	1	<.001	.981	.971	.991
BMI_capped_transformed	.475	.005	9905.303	1	<.001	1.607	1.593	1.623
[HighBP=.000]	-.730	.009	6002.876	1	<.001	.482	.473	.491
[HighBP=1.000]	0 ^b	.	.	0	.	.	.	.
[HighChol=.000]	-.587	.009	4209.966	1	<.001	.556	.546	.566
[HighChol=1.000]	0 ^b	.	.	0	.	.	.	.
[Smoker=.000]	-.009	.009	1.058	1	.304	.991	.974	1.008
[Smoker=1.000]	0 ^b	.	.	0	.	.	.	.
[PhysActivity=.000]	.025	.010	6.100	1	.014	1.025	1.005	1.046
[PhysActivity=1.000]	0 ^b	.	.	0	.	.	.	.
[Fruits=.000]	.054	.009	32.986	1	<.001	1.055	1.036	1.075
[Fruits=1.000]	0 ^b	.	.	0	.	.	.	.
[Veggies=.000]	.057	.011	26.325	1	<.001	1.059	1.036	1.082
[Veggies=1.000]	0 ^b	.	.	0	.	.	.	.
[GenHlth=1.000]	-2.124	.027	6059.336	1	<.001	.120	.113	.126
[GenHlth=2.000]	-1.423	.023	3787.682	1	<.001	.241	.230	.252
[GenHlth=3.000]	-.692	.022	999.246	1	<.001	.501	.480	.523
[GenHlth=4.000]	-.224	.021	115.452	1	<.001	.799	.767	.832
[GenHlth=5.000]	0 ^b	.	.	0	.	.	.	.
[DiffWalk=.000]	-.141	.012	129.908	1	<.001	.869	.848	.890
[DiffWalk=1.000]	0 ^b	.	.	0	.	.	.	.
[Age=1.000]	-1.874	.062	911.313	1	<.001	.153	.136	.173
[Age=2.000]	-1.965	.052	1438.733	1	<.001	.140	.127	.155
[Age=3.000]	-1.626	.038	1849.202	1	<.001	.197	.183	.212
[Age=4.000]	-1.234	.030	1662.594	1	<.001	.291	.274	.309
[Age=5.000]	-.946	.026	1276.571	1	<.001	.388	.369	.409
[Age=6.000]	-.696	.024	877.447	1	<.001	.498	.476	.522
[Age=7.000]	-.504	.021	574.433	1	<.001	.604	.580	.630
[Age=8.000]	-.385	.020	376.947	1	<.001	.681	.655	.708
[Age=9.000]	-.140	.019	54.922	1	<.001	.869	.837	.902
[Age=10.000]	.055	.019	8.687	1	.003	1.057	1.019	1.096
[Age=11.000]	.145	.020	55.029	1	<.001	1.156	1.112	1.201
[Age=12.000]	.122	.021	33.171	1	<.001	1.129	1.084	1.177
[Age=13.000]	0 ^b	.	.	0	.	.	.	.
[Education=1.000]	.101	.145	.484	1	.487	1.106	.833	1.470
[Education=2.000]	.289	.032	81.566	1	<.001	1.335	1.254	1.421
[Education=3.000]	.164	.023	51.755	1	<.001	1.178	1.127	1.232
[Education=4.000]	.095	.012	62.387	1	<.001	1.100	1.074	1.126
[Education=5.000]	.092	.011	64.540	1	<.001	1.096	1.072	1.121
[Education=6.000]	0 ^b	.	.	0	.	.	.	.
[Income=1.000]	.315	.024	179.187	1	<.001	1.370	1.308	1.434
[Income=2.000]	.342	.021	261.221	1	<.001	1.408	1.351	1.468
[Income=3.000]	.274	.019	205.617	1	<.001	1.315	1.267	1.365
[Income=4.000]	.289	.018	271.216	1	<.001	1.335	1.290	1.382
[Income=5.000]	.178	.016	122.525	1	<.001	1.195	1.158	1.234
[Income=6.000]	.122	.014	71.641	1	<.001	1.130	1.098	1.162
[Income=7.000]	.101	.014	54.672	1	<.001	1.107	1.077	1.137
[Income=8.000]	0 ^b	.	.	0	.	.	.	.

a. The reference category is: 0.0.

b. This parameter is set to zero because it is redundant.

By synthesizing the results from the C&RT Decision Tree (which identifies key decision splits), the Random Forest model (which provides a robust feature importance ranking), and the Logistic Regression model (which quantifies risk via odds ratios), a multi-faceted and reliable understanding of the factors influencing diabetes can be established.

Tier 1: Core and Decisive Risk Factors

These factors consistently demonstrated the strongest influence across all three models, acting as the primary drivers of diabetes prediction.

General Health (GenHlth): This emerged as one of the most powerful predictors. It was ranked as the **#1 most important predictor** by the C&RT Decision Tree and the **#3 most important** by the Random Forest model. The Logistic Regression model quantifies this relationship, showing that as self-reported health worsens, the odds of having diabetes increase significantly. This indicates that an individual's overall perception of their health is a critical comprehensive indicator of diabetes risk.

Body Mass Index (BMI): BMI is another core factor. It was identified as the **#1 most important predictor** by the Random Forest model (F1) and the **#3 predictor** by the C&RT model. The Logistic Regression provides a clear quantitative risk: the odds ratio ($\text{Exp}(B)$) for BMI_capped_transformed is 1.607. This means that for every one standard deviation increase in BMI, the odds of an individual having diabetes increase by approximately 60.7%, holding all other factors constant.

Age: The relationship between increasing age and diabetes risk is prominent. Age was the **#2 most important predictor** in the Random Forest model (F2). The Logistic Regression model confirms this strong trend, with odds ratios progressively increasing for older age groups, indicating a significantly higher risk compared to the youngest cohort (18-24 years).

High Blood Pressure (HighBP): As a key clinical indicator, HighBP was ranked **#2** by the C&RT model and **#5** by Random Forest (F5). The Logistic Regression model shows that individuals with high blood pressure have significantly higher odds of having diabetes compared to those without.

Tier 2: Significant Contributing Factors

These factors have a clear and significant impact on diabetes risk, though they rank slightly below the core factors in the predictive models.

High Cholesterol (HighChol): This factor showed moderate importance in both the C&RT and Random Forest models. The Logistic Regression confirms its status as a significant risk factor, with an odds ratio indicating that individuals with high cholesterol have substantially higher odds of having diabetes.

Socioeconomic Factors (Income & Education): The Random Forest model highlighted the strong predictive power of these factors, ranking Income as **#4** (F4) and Education as **#7** (F7) in importance. The coefficients from the Logistic Regression model support this, showing a trend where lower levels of

income and education are associated with higher odds of diabetes after controlling for other variables.

Physical Health (PhysHlth) and Difficulty Walking (DiffWalk): Indicators of physical well-being, such as the number of recent days with poor physical health (PhysHlth_transformed) and having difficulty walking, consistently appeared as significant predictors across the models.

Factors with Minor or Ambiguous Influence

These factors, while statistically significant in some cases, demonstrated a much smaller predictive influence compared to the factors listed above.

Daily Lifestyle Behaviors (Fruits, Veggies, PhysActivity, Smoker): These variables consistently ranked in the lower half of the feature importance charts for both Random Forest and the C&RT model. While the Logistic Regression shows they are statistically significant, their odds ratios are generally close to 1.0 (e.g., PhysActivity=1.000 has an odds ratio of 1.025). It is likely that the effects of these behaviors are largely mediated or "absorbed" by the more dominant factors like BMI and General Health. For example, a poor diet and lack of exercise directly contribute to a higher BMI, making BMI the more direct and powerful predictor in the models.

Mental Health (MentHlth): This variable consistently showed low-to-moderate importance across all models. Its odds ratio in the Logistic Regression model is 0.956, which is very close to 1.0, suggesting a statistically significant but very small protective effect on its own.

Conclusion

The synthesized analysis across the C&RT, Random Forest, and Logistic Regression models robustly identifies **self-perceived general health, BMI, age, high blood pressure, lower socioeconomic status(Income & Education) as the 5 most influential predictors of diabetes**. Following these are other significant contributors including high cholesterol and general physical functioning. In contrast, while specific daily lifestyle choices like diet and smoking are undoubtedly important for health, their direct predictive influence in the models is less pronounced, likely because their impact is captured by the more dominant physiological and demographic factors.

8.3 Cluster Analysis for Health Profile Segmentation

To add further depth to the classification results and identify actionable population

segments, a supplementary cluster analysis was performed. Crucially, the Diabetes_binary target variable was excluded from the clustering model-building process (role was set to 'None'). This ensures that the resulting segments are formed based purely on the participants' behavioral, physiological, and demographic profiles, without any prior knowledge of their diabetes status. Using the K-means algorithm, five distinct health profiles (clusters) were identified from the dataset. The resulting model had a Silhouette measure of 0.213, indicating a weak but discernible separation of profiles. The fact that these independently-formed clusters show a significant variation in diabetes rates strongly validates that these personas are meaningfully associated with diabetes risk. This analysis provides insights into how different risk factors combine, which can be used to design targeted public health interventions.

Cluster	Record Count	Diabetes (Mean)	HighBP (Mean)	HighChol (Mean)	Smoker (Mean)	PhysActivity (Mean)	BMI (Mean)	Age (Mean)	Income (Mean)	GenHlth (Mean)
1	46,627	0.242	1.000	0.998	0.479	0.782	0.15	9.48	6.07	2.64
2	70,314	0.056	0.186	0.207	0.000	0.878	-0.21	7.26	6.65	2.06
3	36,826	0.340	0.754	0.628	0.644	0.394	0.49	9.38	4.24	3.86
4	54,999	0.075	0.219	0.240	1.000	0.887	-0.17	7.83	6.32	2.26
5	44,914	0.077	0.208	0.226	0.254	0.677	-0.01	6.89	6.27	2.29

(Note: BMI, Age, Income, and GenHlth are presented as mean scale values for comparison across clusters.)

Interpretation of Health Profiles

Cluster 1: Cardiovascular High-Risk Group: Defined by nearly universal rates of high blood pressure and high cholesterol, with a high diabetes prevalence of 24.2%.

Cluster 2: Low-Risk Healthy Group: The healthiest segment, with the lowest diabetes rate (5.6%), non-smokers, and physically active.

Cluster 3: High-Risk Diabetic & Lifestyle Group: This cluster has the highest prevalence of diabetes at 34.0%, associated with smoking, low physical activity, and high BMI.

Cluster 4: "High-Activity Smokers" Group: Composed entirely of smokers but with high physical activity levels and a below-average diabetes risk (7.5%).

Cluster 5: Poor Dietary Habits Group: Characterized by poor diet and moderate physical activity, with a low current diabetes risk (7.7%) but a clear target for preventative campaigns.

Implications

This segmentation provides a clear roadmap for targeted public health strategies. Rather than a one-size-fits-all approach, campaigns can be tailored:

For Cluster 3: Interventions must be intensive and address multiple fronts: smoking cessation, promoting accessible physical activity, and weight management support.

For Cluster 1: The focus should be on integrated care, managing blood pressure and cholesterol to prevent diabetes and its complications.

For Cluster 4: Campaigns can be designed to appeal to their active lifestyle, framing smoking as a barrier to achieving peak physical performance.

8.4 Evaluation of Results, Models and Patterns

The evaluation of the six models across the 30% holdout test set yielded several clear insights. As summarized in the performance table below, the application of the oversampling directive had a profound impact on model behaviour. For the C&RT Decision Tree and Logistic Regression models, it dramatically increased Recall for the diabetic class (from under 15% to over 74%), successfully meeting the project's primary success criterion of $\geq 75\%$ recall. This critical improvement in detecting positive cases came at the expected cost of overall accuracy and a significant drop in Precision, demonstrating the classic trade-off required to mitigate the model's inherent bias towards the majority class.

Among the three algorithms, Random Forest emerged as the unequivocally superior model. It achieved the highest AUC (≈ 0.956) in both balanced and imbalanced scenarios, indicating the best overall discriminative power. Furthermore, when trained on the SMOTE-balanced data, the Random Forest model was the only one to achieve a strong and balanced performance across both Recall (78.6%) and Precision (75.7%), making it the most robust and reliable model for this task.

Model	Training Data	Accuracy	Recall (Diabetic)	Precision (Diabetic)	AUC
C&RT Decision Tree	SMOTE Balanced	72.05%	74.70%	29.90%	0.784
Random Forest	SMOTE Balanced	93.50%	78.60%	75.70%	0.955
Logistic Regression	SMOTE Balanced	72.23%	76.80%	30.40%	0.82
C&RT Decision Tree	Imbalanced	86.46%	11.70%	56.80%	0.722
Random Forest	Imbalanced	94.83%	70.30%	90.40%	0.955
Logistic Regression	Imbalanced	86.52%	14.80%	56.10%	0.819

This multi-metric approach ensures that models are not only assessed on overall correctness but also on their ability to handle class imbalance. Non-SMOTE models were biased towards the majority class, achieving high accuracy but poor recall, which is problematic in medical applications where missing positive cases can have serious consequences. In contrast, SMOTE-based models demonstrated improved recall and F1-scores, validating the effectiveness of resampling techniques for imbalanced datasets. Cross-model comparison highlights Random Forest (SMOTE)

and Logistic Regression (SMOTE) as the most effective models, striking a balance between interpretability, sensitivity, and overall predictive performance.

8.5 Multiple Iterations and Model Refinement

Several iterations were conducted to improve robustness and effectiveness.

Iteration 1 - Baseline Models: Initial models (Decision Tree, Random Forest, Logistic Regression) were trained on the non-SMOTE dataset to establish baseline performance.

Iteration 2 - Feature Engineering: Variable transformations (e.g., BMI capped transformation, age_group) and the removal of weaker predictors were tested to improve signal strength.

An initial hypothesis to create a composite Health_Index was tested by deriving the new variable. However, subsequent correlation analysis demonstrated that this new feature was less predictive than its individual components and introduced redundancy. Based on this evidence, the feature was discarded in a subsequent iteration, and the original variables were retained.

Continuous variables such as BMI_capped, PhysHlth, and MentHlth are measured on vastly different scales. BMI ranges from approximately 10 to 60, while PhysHlth and MentHlth range from 0 to 30. This disparity in scale can bias algorithms that rely on distance calculations or gradient descent, as a unit change in BMI would have a much larger impact than a unit change in PhysHlth. Hence the Z-score normalization was applied to all continuous predictor variables (BMI_capped, PhysHlth, MentHlth).

Iteration 3 - Algorithm Selection: During the modeling phase, an incompatibility was discovered between the initially chosen CHAID algorithm and the synthetic data generated by the oversampling directive, resulting in a model failure. The process was iterated by diagnosing the issue and replacing CHAID with the more robust C&RT algorithm, which successfully handled the data and produced an interpretable model.

Iteration 4 – Clustering Exploration: Cluster analysis was performed to better understand data distribution and potential hidden group structures, informing variable selection.

Iteration 5 – SMOTE Implementation: SMOTE was applied to rebalance the dataset, addressing class imbalance and enhancing minority class detection. Models were retrained and evaluated on this new dataset.

Iteration 6 – Model Comparison and Refinement: Results from both SMOTE and non-SMOTE runs were systematically compared. Logistic Regression and Random Forest with SMOTE emerged as the most effective approaches.

This iterative process demonstrates systematic refinement and validation, ensuring that the final conclusions are based on robust experimentation rather than a single modelling attempt.

References

American Diabetes Association. (2023). *Economic Costs of Diabetes in the U.S. in 2022*. <https://diabetes.org/about-diabetes/statistics/about-diabetes>

International Diabetes Federation. (2025). *IDF Diabetes Atlas (10th ed.)*. <https://diabetesatlas.org/>

Centers for Disease Control and Prevention. (2024). *National Diabetes Statistics Report*. U.S. Department of Health and Human Services. <https://www.cdc.gov/diabetes/php/data-research/index.html>

Sortsø, C., Komkova, A., Sandbæk, A., Griffin, S. J., Emneus, M., Lauritzen, T., & Simmons, R. K. (2018). Effect of screening for type 2 diabetes on healthcare costs: a register-based study among 139,075 individuals diagnosed with diabetes in Denmark between 2001 and 2009. *Diabetologia*, 61(6), 1306–1314. <https://doi.org/10.1007/s00125-018-4594-2>

Academic Integrity Disclaimer

"I acknowledge that the submitted work is my own original work in accordance with the University of Auckland guidelines and policies on academic integrity and copyright. (See: <https://www.auckland.ac.nz/en/students/forms-policies-and-guidelines/student-policies-and-guidelines/academic-integrity-copyright.html>).

I also acknowledge that I have appropriate permission to use the data that I have utilised in this project. (For example, if the data belongs to an organisation and the data has not been published in the public domain then the data must be approved by the rights holder.) This includes permission to upload the data file to Canvas. The University of Auckland bears no responsibility for the student's misuse of data."